Information Economics and Policy 71 (2025) 101134

Contents lists available at ScienceDirect

Information Economics and Policy

journal homepage: www.elsevier.com/locate/iep

Content providers and the deployment of Internet infrastructure ☆

El Hadi Caoui, Andrew Steck ,*

U of Toronto, Canada

ARTICLE INFO

JEL classification:
L13
L96
L86

Keywords:
Internet backbone
Content providers
Bandwidth
Vertical integration

ABSTRACT

This paper documents the growing role that content providers play upstream in the global internet supply chain. Using novel data, we establish three stylized facts: (1) As bandwidth buyers, content providers' share of global used bandwidth has grown dramatically (from 5% to 69%) over the period 2005–2021, with important heterogeneity across regions. (2) Content providers (in particular, Amazon, Google, Meta, and Microsoft) account for an increasingly large share of investment in internet cable infrastructure. (3) The growth in content providers' demand is associated with the roll-out of their data centers globally and corresponding increase in inter-data center traffic; their investment in private cables is in part driven by data center siting, which are in locations that may lack connectivity to public internet cables. We discuss implications of these trends for innovation, internet traffic transparency, technology standard adoption, and network resilience.

1. Introduction

In recent years, much attention has been focused on the growing importance of large technology companies in digital markets. Recent examples on the regulatory side include the Digital Markets Act in the EU and its stated intent to ensure fair market access and enhance market contestability (European Parliament (2023)). Antitrust agencies in the US and EU have also directed their attention toward potential abuses of dominant market positions by technology firms.[1] However, this attention and the existing research on the economics of technology platforms have been restricted to the consumer-facing side of the internet: see, for example, recent work on self-preferencing and platform bias (Farronato et al. (2023), Reimers and Waldfogel (2023)). By contrast, this paper documents trends in the role played by large technology firms in the upstream segment of the internet's vertical chain: we highlight their role as large buyers in bandwidth markets and more recently—via vertical integration—as owners of an increasing share of internet infrastructure.

Our analysis focuses on a category of firms that we refer to as "content providers". This category includes large providers of internet content, such as social media platforms and video streaming sites (Google, Meta, Microsoft, and Apple), cloud service providers (Amazon, Microsoft, Google, Oracle, OVH), and network-based content delivery networks (CDNs) (such as Akamai and CloudFlare).[2] We distinguish content providers from internet backbone providers; the latter are telecommunications carriers that operate public networks and carry internet traffic (e.g., Verizon, Orange, AT&T, Vodafone).[3] As we document in

☆ We thank Heski Bar-Isaac, Tim Bresnahan, Avi Goldfarb, Jan Tichem, and Joel Waldfogel for helpful discussions, and participants at the Mannheim Center for Competition and Innovation (MaCCI) annual conference for feedback. The authors are grateful to the NET Institute www.NETInst.org for its financial support of this research project. The analysis, findings, and conclusions expressed in this paper should be attributed solely to the authors, neither to TeleGeography nor the NET Institute.

* Corresponding author.
E-mail addresses: elhadi.caoui@rotman.utoronto.ca (E.H. Caoui), andrew.steck@rotman.utoronto.ca (A. Steck).

[1] See *United States, State of Arkansas, State of Florida, State of Georgia, State of Indiana, Commonwealth of Kentucky, State of Louisiana, State of Mississippi, State of Missouri, State of Montana, State of South Carolina and State of Texas v. Google LLC* and *Federal Trade Commission v. Meta Platforms, Inc.*

[2] A Content Delivery Network (CDN) is a network of geographically distributed servers and data centers designed to enhance the delivery of web content to end-users (Hau et al. (2011), Chiang and Hua Jhang-Li (2014)). CDNs strategically place copies of content in multiple server locations around the world. When a user requests content, the CDN's system redirects the request to the nearest server (usually based on the user's geographic location), rather than the origin server where the content is originally hosted.

[3] Internet backbone providers, or Tier-1 providers, are a type of internet service provider (ISP) that has access to the entire domain of the internet solely via peering agreements with other backbone providers, meaning they do not have to purchase internet transit from other providers. These ISPs have their own high-capacity networks and form the core of the internet backbone.

https://doi.org/10.1016/j.infoecopol.2025.101134
Received 28 March 2024; Received in revised form 15 April 2025; Accepted 9 May 2025
Available online 16 May 2025
0167-6245/© 2025 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

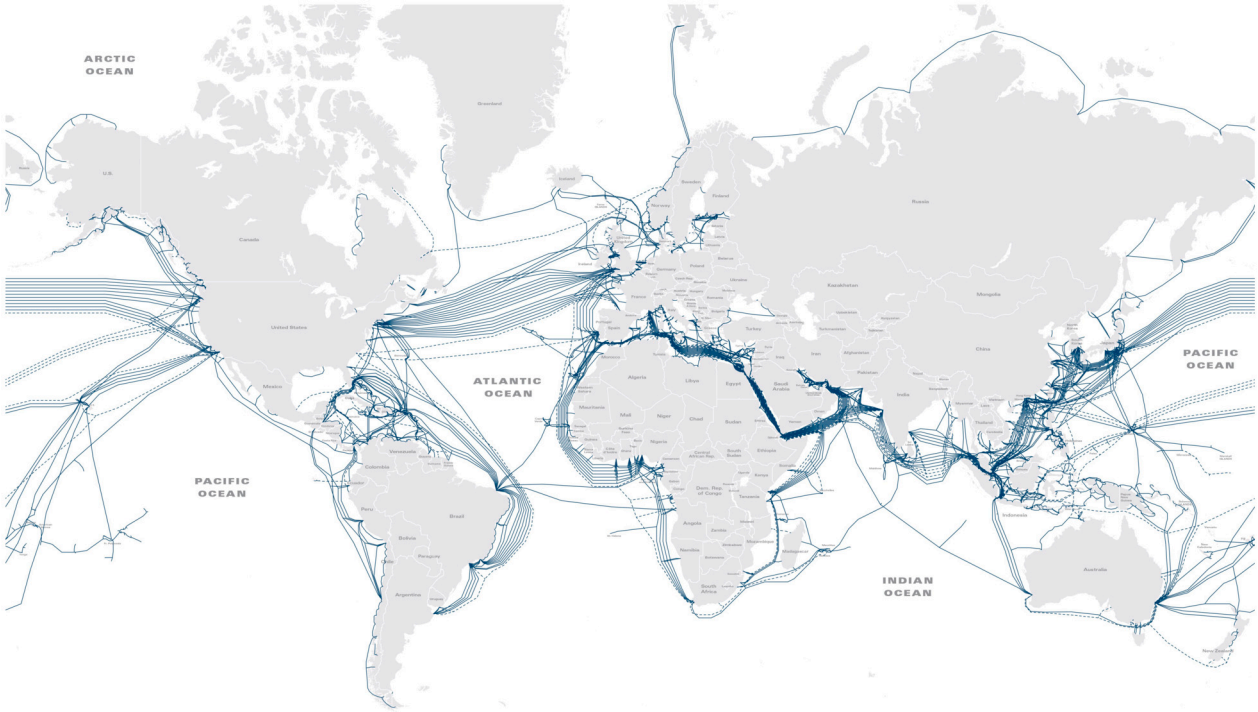


Fig. 1. The Global Internet Backbone (source: TeleGeography).

detail below, the historical decoupling of the market for internet backbone carriage and the market for content provision has diminished in recent years.

We study the role played by content providers in the upstream segment of the internet supply chain, focusing on global bandwidth markets and the infrastructure supporting them (i.e., the so-called "internet backbone") over the period 2005 to 2021. An important facet of internet infrastructure involves the global network of undersea fiber-optic cables (see Fig. 1). This network serves as the primary mean for intercontinental digital exchanges, encompassing over 98% of all international internet traffic, including data transmissions, video conferencing, messaging, and emails (Brake (2019)). The operational undersea cable network plays a pivotal role in sustaining the operation of global economic and financial systems. For instance, the U.S. Clearing House Interbank Payment System channels in excess of $10 trillion in daily financial transactions, connecting with more than 22 economies through undersea cables (Federal Communications Commission (2015), Sunak (2017)).[4]

Our study leverages comprehensive and novel data from TeleGeography, a market research firm specializing in the telecommunications industry. TeleGeography's data offers comprehensive information on: (1) undersea cable systems, including ownership, landing points, entry dates, design capacities, length, and construction costs. These data allow us to track the involvement of content providers in the deployment of undersea cable infrastructure. (2) Annual panels of used bandwidth (or data flows) at various geographic levels, e.g. by country-pair or region-pair (e.g., North America–Europe). Used bandwidth at the region-pair level is broken down by the type of buyer deploying that bandwidth (i.e., content provider, internet backbone provider, enterprise, research and educational). These data allow us to measure the share of international bandwidth demand accounted for by content providers over time and by region. (3) Annual panels on data center roll-out by Amazon, Google, Meta, and Microsoft at the city level. These data allow us to examine the role data center infrastructure plays in shaping content providers' demand and investment in undersea cable infrastructure. Taken together, these data offer new insights into the growing importance of content providers; we illustrate these insights via three stylized facts.

First, our analysis shows that content providers play an increasingly important role as buyers of bandwidth. From 2005 to 2021, the global share of international bandwidth used by content providers has increased from around 5% to 69%. In 2017, content providers surpassed internet backbone providers as the largest users of international capacity, and this lead has continued to rapidly increase in subsequent years. Notably, between 2017 and 2021, the deployment of international bandwidth capacity by content providers surged over six-fold, reaching 2 Pbps (or 2,000 Tbps). We document the heterogeneity in these trends across routes (e.g., Trans-Pacific, Trans-Atlantic, Europe-Asia).

Second, content providers have become important investors in internet infrastructure. While content providers have been important buyers of international bandwidth for nearly a decade, they have not historically been investors in new cables, particularly in terms of capital expenditure for new construction. However, the landscape of investment in new cables has seen a marked shift in recent years. Content providers increasingly operate private long-haul networks that are distinct from the public internet networks operated by internet backbone providers. This strategy of vertical integration into cable operations and investment in new (and often private) cable systems has emerged in the last few years and is expected to increase in the coming years.

This trend began when Google became an investor in the Unity cable consortium, which entered service in 2010 (Gardiner (2008)). Subsequently, major players such as Meta, Microsoft, and Amazon have also made direct investments in new undersea cable systems: the share of new cables with a content provider investment has risen from 3.5% of new cables from 2010 through 2015, to 24.4% of new (and projected) cables from 2021 through 2026. We detail the changing patterns of these investments over the past decade (and projections for the next few years) and compare them to investments made by internet backbone providers. We show that on the major routes such as the Trans-Atlantic and Trans-

[4] We do not consider the network of terrestrial cables, arguably another important component of the internet backbone in which content providers are increasingly active, due to a lack of data on terrestrial infrastructure.

Pacific, the majority of new and planned cables are either solely-owned or invested in by content providers.

Third, the growing role of content providers as both bandwidth buyers and cable operators is driven by their specific demand. Content providers' demand can be categorized into two segments: inter-data center demand and content distribution and cloud services (Tan (2023)). Inter-data center demand refers to the high volumes of data that are transferred between different data centers owned by the same company, encompassing activities such as database mirroring, search index synchronization, and cloud computing services.[5] Content distribution and cloud services require content providers to connect their data centers with end-users, improving performance and control over content distribution.[6] We leverage data on the roll-out of data centers by major content providers to explore some of the sources explaining the recent upward trends in the share of global used bandwidth attributable to content providers and their investment decisions in cable infrastructure.

The growth in content providers' demand is associated with the roll-out of their data centers globally: in particular, inter-data center traffic is an important driver of aggregate data traffic. We show, via an event study design, that after a content provider starts operating data centers on both ends of a given route (e.g., US-Spain), traffic on that route increases by 170% within 5 years. We also find evidence that content providers invest in internet cable infrastructure to connect their network of data centers: direct investment in cables allows them to shape the design of undersea cable networks to better connect their data centers. Data centers can be deployed in locations that may not be well connected to the existing public internet cable infrastructure because of their smaller size (in terms of bandwidth traffic) or other cost consideration (e.g. corporate taxes).

In light of these stylized facts, we discuss potential implications of these trends in the medium to long-term. In particular, we focus on the potential effects that the growing role played by content providers may have on the quality of services and latency, on innovation and investment in public internet infrastructure, on the adoption of new technology standards, and on the resilience of internet infrastructure.

The primary contribution of this paper is to provide new insights into the growing involvement of content providers in the use and deployment of upstream internet infrastructure. We leverage new data on global internet traffic flows and investment in internet infrastructure (e.g., undersea cables and data centers) to consider the role of content providers from a previously unstudied angle. We adopt a descriptive approach, which we view as a necessary first step in the development of an understanding of content providers' emerging impact in the upstream segment of digital markets and informing future research on the industrial organization of the internet supply chain.

Literature Review. This paper is related to the literature analyzing the implications of net neutrality regulation. Net neutrality refers to the principle that all data packets on an information network are treated equally: in particular, an internet service provider (ISP) may not charge fees to content providers for accessing users and may not discriminate (in terms of price or quality of transmission) across different content providers. Schuett (2010), Faulhaber (2011), Krämer et al. (2013), Greenstein et al. (2016), Easley et al. (2018), and Jamison (2019) survey the literature on net neutrality and discuss remaining open questions. Hildebrandt and Wiewiorra (2024) provides an updated overview which also connects the net neutrality concept to the growing role of digital platforms and content providers.

On the theoretical front, the literature has analyzed various aspects of net neutrality: e.g., termination fees charged by ISPs to content providers and the zero-price rule (Musacchio et al. (2009), Lee and Wu (2009), Economides and Tåg (2012), Njoroge et al. (2013), Gans (2015), Gans and Katz (2016)), non-discrimination, prioritization and congestion issues (Hermalin and Katz (2007), Choi and Kim (2010), Becker et al. (2010), Cheng et al. (2011), Krämer and Wiewiorra (2012), Economides and Hermalin (2012), Bourreau et al. (2015)), and incentives for vertical foreclosure (Chen and Nalebuff (2006), Economides (2008), and Broos and Gautier (2017)). One strand of the empirical literature estimates the effect of net neutrality regulations on network investment, innovation, or social welfare: e.g., Hazlett and Wright (2017), Ford (2018), Hooton (2020) for the U.S., Lee and Kim (2014) for South Korea, and Layton (2017) and Briglauer et al. (2023) for the EU. These contributions generally find a negative impact of such regulations on network investment. Another strand focuses on foreclosure and gatekeeping: in the case of over-the-top video (OTT), McManus et al. (2023) and Malone et al. (2023) study an ISP's incentive to steer consumers toward their TV service by either increasing the price of internet service or degrading the access to and quality of streaming content; whereas McManus et al. (2024) analyze the role of usage based-pricing in steering consumers from streaming video to traditional TV subscriptions sold by the ISP.

The net neutrality literature focuses on the behavior of ISPs in the "last-mile" or residential access market. By contrast, we consider (Tier 1) internet providers in the upstream backbone market.[7] Nonetheless, several concerns raised in the domestic net neutrality debate (e.g., vertical foreclosure) may also be relevant when considering vertical integration by content providers at the inter-continental level.

Relatedly, the stylized facts we highlight are connected to recent and potential disputes in interconnection arrangements between ISPs and content providers (including CDNs). Internet interconnection is largely an unregulated aspect of internet infrastructure, meaning that ISPs are not required to connect with every content provider, nor are the terms of these agreements regulated or publicly disclosed. Crémer et al. (2000), Laffont et al. (2001), Besen et al. (2001), and Laffont et al. (2003) provide early theoretical treatment of peering between ISPs and (Tier-1) backbone providers. Typically, ISPs agree to settlement-free peering arrangements when the ratio of inbound to outbound traffic is roughly balanced (Wang et al. (2021)). If the traffic is not balanced, content providers may negotiate bilateral peering agreements with ISPs, which involve direct payments (Little and Wright (2000), Coucheney et al. (2014), Choi et al. (2015), Kim (2020)).

Disputes between content providers and ISPs over settlement-free peering often arise due to disagreements about traffic imbalance and the associated costs (Clark et al. (2011), Chang et al. (2006), and Nikkhah and Jordan (2023)). When content providers generate a large volume of downstream traffic, ISPs may feel that they bear the brunt of the infrastructure costs without adequate compensation, leading them to seek payments or altered terms. Content providers, on the other hand, may assert that carrying their traffic to an interconnection point close to consumers is of value, that their content drives user demand, benefiting ISPs by attracting subscribers. This misalignment in perceived value and cost-sharing responsibilities frequently leads to conflicts regarding maintaining settlement-free peering arrangements and the sharing of network costs. For example, in 2014, Netflix accused Comcast and Verizon of strategic under-provisioning of interconnection capacity in order

[5] Database mirroring is a technique used to create redundant copies of a database to ensure data availability and reliability. Search index synchronization refers to the process of keeping search indices up to date across different servers or data centers. A search index is a data structure that improves the speed and efficiency of search queries. Cloud computing services are services delivered over the internet that provide computing power, storage, and various applications on demand. Instead of owning and maintaining physical hardware, organizations can use cloud services to access resources such as virtual machines, storage, databases, and software applications, often on a pay-as-you-go basis.

[6] The bandwidth requirements for content distribution tend to be smaller than for inter-data center traffic, but a larger number of locations must be connected, as opposed to a smaller number of large data centers.

[7] While content providers have also increased their involvement in the deployment of terrestrial backbone networks, limited data exists on this infrastructure, hence our focus on the inter-continental backbone.

to degrade the quality of experience of their subscriber base. Goetz (2019) models the dynamic bargaining between Netflix and last-mile ISPs over the terms of the interconnection agreement.

Our paper sheds light on several aspects relevant to peering disputes: e.g., we leverage new data that supports the claim that content providers account for the majority of used bandwidth, in particular on the global backbone network. We also provide evidence that a significant fraction of global traffic now consists of data that is moved between the data centers of large content providers. In addition, recent work on peering (Nikkhah and Jordan (2023)) show that private investment in infrastructure by large content providers, which we document in the case of undersea infrastructure, may facilitate peering negotiations with ISPs by ensuring that content is carried on the content provider's network to interconnection points close to the end user, alleviating ISP's network costs.

This paper also contributes to the literature studying the deployment of internet infrastructure. Empirical work in this strand has analyzed aspects related to investment in terrestrial broadband infrastructure: e.g., the impact of net neutrality on investment (see references above), the role of access regulation and its impact on investment (Grajek and Röller (2011), Briglauer (2015), Briglauer et al. (2018)), entry delays and preemption in U.S. broadband investment (Skiti (2020), Wilson et al. (2021), Kang and Xiao (2023)), the role of public policy in broadband deployment (Briglauer et al. (2014), Wilson (2016), Briglauer et al. (2020), Wilson (2023a)), and the effect of competition and multimarket contact on quality of service (Wilson (2023b)). Greenstein (2015) provides a comprehensive examination of the internet's commercialization stages, including broadband investment.

Several studies examine the economy-wide effects of undersea internet cables. Hjort and Poulsen (2019) shows that the introduction of the first undersea internet cables to various African regions significantly increases local employment, wages, and firm productivity. Cariolle et al. (2024) finds that expanding undersea cables reduces the price of internet access. Simione and Li (2021) and Anderson and O'Connor (2020) report that cable arrivals in Africa are associated with higher GDP per capita, labor productivity, and employment in fiber-connected areas. Imbruno et al. (2022) and Cariolle and da Piedade (2023) investigate the impact of digital connectivity via undersea cables on export markets, finding positive effects.[8] Fanou et al. (2020) analyzes the impact of a major cable investment between South America and Africa in 2018, finding it broadly reduced latency between the two regions. In light of our findings regarding the role of data center traffic and location, it is *a priori* unclear whether investments in undersea cables by content providers would yield the same economy-wide effects as those by backbone providers.[9]

Two recent economic studies have examined entry and competition in the undersea internet cable industry. The first paper, Caoui and Steck (2024), uses undersea cable disruptions as a case study of the effect of supply disruption on market structure. In this industry, buyers commonly manage disruption risk (from cable failures) by provisioning bandwidth from multiple competing cables. Caoui and Steck (2024) analyze the effect of this diversification on new cables' incentive to enter and the nature of competition between cable operators. Using a structural model combining demand for internet bandwidth across regions and cable entry and bandwidth supply choices, Caoui and Steck (2024) show that supplier diversification drives a significant portion of new cable investments and surplus generation. Caoui and Steck (2024) focuses on cable disruptions and abstracts from any specific role played by content providers. By focusing on content providers, this paper complements the previous study by directing attention to recent changes in undersea cable ownership and investment, which has received less attention in the literature.

The second paper, Jeon and Rysman (2024), proposes a model for the routing of global data flows and quantifying network externalities from new undersea cable entry across regions. The paper also decomposes growth in internet usage into growth in demand and improvements in the cable network, finding that cable deployment is an important contributor to bandwidth growth. In contrast to Caoui and Steck (2024), Jeon and Rysman (2024) allow content providers and backbone providers to differ in their objective functions when making cable entry decisions. They interpret the difference in payoffs from building a new cable as indicative that content providers have different motives for building cables, such as connecting their own data centers. The evidence we present here with the third stylized fact—linking cable landing decisions and the presence of data center infrastructure—supports their interpretation.

Datacenters, another important segment in the internet supply chain, have recently been the subject of empirical analysis: e.g., Greenstein and Pan Fang (2020) study entry strategies used by third-party data centers in the United States. Demirer et al. (2023) study the role of data storage and computation in production using data from a large cloud provider and estimate the costs of data regulation.

The rest of the paper proceeds as follows: Section 2 provides background on the empirical setting and details on the data used. Section 3 documents the role content providers play as buyers in international bandwidth markets, and Section 4 documents the growing involvement of content providers in the deployment of internet infrastructure. Section 5 analyzes the role that data center investment plays in driving both content providers' demand and their decisions to invest in undersea cables. Section 6 discusses potential implications of these trends in the medium to long-term, and Section 7 concludes.

2. Industry background and data sources

This section contains two parts. First, it provides an overview of the industry under study. Second, it introduces and describes the data sources we use.

2.1. Industry background

The infrastructure (physical and organizational) that enables the operation of the internet is a complex, multilayered, and famously decentralized landscape. Some of the layers include retail ISPs and their networks, internet exchange points, telecommunications and infrastructure investment firms, physical cables (fiber-optic or other), data centers, websites (commercial and other), operating systems, protocols and browsers, cellular towers and satellites, and finally the servers, computers, and mobile devices which use these networks to connect and send/receive information. As detailed in Section 1, facets of many of these layers have been the object of academic studies.

In this paper, we focus attention on the largest internet firms, the content providers, and their relationship with a fundamental layer of internet infrastructure: the network of undersea fiber-optic cables. Despite the importance of this component of internet infrastructure (e.g., carrying 98% of international internet traffic), it has received little attention from academic economists.[10]

Undersea fiber-optic cables are only the most recent iteration in the centuries-old market in inter-continental communication. The "packet

[8] In a related context, Steinwender (2018), following Hoag (2006) and Garbade and Silber (1978), shows that early transatlantic telegraph cables strengthened links between markets across the Atlantic and improved trade flow efficiency.

[9] Anderson et al. (2021) estimates the economic impact of Meta's investment in the MAREA undersea cable by combining estimates of cable arrival effects on broadband prices with measures of how broadband prices and penetration affect GDP.

[10] The two recent exceptions are Caoui and Steck (2024) and Jeon and Rysman (2024).

trade" of regularly scheduled mail-carrying ships was succeeded in the mid-late 19th century by transoceanic telegraph cables. Telephone lines (originally copper) followed next in the mid 20th century, with fiber-optic cables (originally carrying phone calls) arriving on the scene near the end of the 20th century (Swinhoe, 2021).

Contemporary undersea fiber-optic cables feature multiple concentric layers, including several glass optical fibers at their core which transmit data via signals of light. Surrounding the core are protective sheaths and a copper layer, which carries electrical power. These cables, generally 25 mm or so in diameter, are typically thicker for the portions that lie in shallower waters due to the addition of extra protective layers. To combat signal degradation, repeaters or optical amplifiers are installed at intervals of 60-100 km. Landing stations at both ends manage the transfer of data between the cable and data centers, internet exchange points, or the local networks.

The cost of building a new cable varies, depending primarily upon the length of the cable. With the cost of the cable itself ranging from $10 to $25 thousand dollars per kilometer and auxiliary equipment such as repeaters, landing stations, etc. representing a cost from $1 to $10 million, the all-in costs of constructing a new cable is on average in the hundreds of millions of dollars. The industry typically has no economic regulation to speak of, and subsidies are not generally provided to offset construction costs.

Globally, the ownership of undersea cables is not concentrated: our data features 400 distinct owners with ownership stakes in 434 cables. The typical owner of such cables has historically been a major telecommunications firm and less-typically a government or regional telecommunications firm.[11] These telecommunications firms were selling connectivity but not competing in the content provision space. As of this writing, content providers have begun shifting to making infrastructure investments, either through building undersea cables directly, or by investing in consortia doing so. This trend is examined in detail in Section 4.

Cable bandwidth is sold wholesale by cable owners to a diverse set of clients. These include internet backbone providers (Tier-1 ISPs), content providers, private businesses, and research/educational bodies. Internet backbone providers are the legacy telecommunications firms that have sold access to the internet since its earliest days. These firms will either own and operate their own cables, or lease capacity on other firms' cables, or both. Content providers are made up of the largest tech firms, as well as legacy media companies. Most bandwidth demanded comes from customers in these first two categories. Private and research entities, the remaining two categories, together comprise a relatively small share of bandwidth demand.

This paper does not directly study or use bandwidth prices (a panel of price samples is available in a different TeleGeography dataset), but we briefly discuss recent market dynamics here for context. Quoted prices for a representative 1-year lease of bandwidth (at 10 Gbps) have been falling over the time-frame studied. As detailed in Caoui and Steck (2024), the median price fell from $55.5 thousand per month to $2.2 thousand per month from 2005 to 2021. However, significant price dispersion remains: the 25th and 75th percentile quotes for 2021 are $1.1 and $12.0 thousand per month respectively.

There is also significant dispersion across the size of markets, ranging from those on the busy Trans-Atlantic and Trans-Pacific routes, to those on the relatively smaller Africa-Asia route. Fig. A.1 plots the bandwidth demanded by region over time, providing a direct illustration of the relative market sizes in this global market.

In recent years, the majority of growth in international bandwidth demand has been driven largely by content providers, a phenomenon we examine in detail in Section 3. One oft-cited reason for this trend is a proliferation in the number of content provider-owned data centers and the corresponding rise in inter-data center traffic. For example, in order to reduce latency experienced by customers, Google subsidiary YouTube may regularly ensure that the datafiles for popular videos are copied and housed in data centers globally (an application known as 'content distribution'); similarly, cloud service providers such as Amazon and Microsoft work to make sure that cloud applications and services are located near to end users across geographies.

2.2. Data sources

The primary data source for this paper, consisting of undersea cable characteristics, a panel of international bandwidth traffic, and data center locations are all from TeleGeography, a telecommunications market research firm. We supplement these data with auxiliary country-level statistics (e.g., GDP), electricity generation, electricity prices, and corporate tax rates.

TeleGeography

TeleGeography collects and retains comprehensive datasets on various aspects of the telecommunications industry; in the present study we draw on data contained in TeleGeography's Global Bandwidth Research Service (GBRS). Unless otherwise indicated, all data series are available for the period 2005-2021.

The Global Bandwidth Research Service (GBRS) contains detailed data on over 400 undersea cable systems. The data on each cable system includes: detailed landing points (where the cable comes ashore, connecting to the terrestrial network), the historical (or projected future) date of entry into service, the designed (or potential) capacity (i.e., total bandwidth in Gbps that can be carried by the cable), the cable's length, construction costs, and ownership.[12] These data form the basis of our analysis in Sections 4 and 5. The GBRS dataset also contains data on a number of under-construction and announced future cables, with entry dates ranging from 2022-2026; we include these projected cables in many of our analyses of the evolving role that content providers play on the supply side of the market in Section 4.

In addition to these cable-level details, the GBRS dataset also features an annual panel of used bandwidth. This panel is available at multiple geographic levels, by region-to-region or by country-to-country. These data are used in Sections 3 and 5.

The GBRS dataset also provides information on an important determinant of bandwidth demand and undersea cable supply: the roll-out of data centers owned by some of the largest technology firms. The GBRS data include information on data centers regions owned by Amazon, Google, Meta, and Microsoft, who represent the largest firms operating in the cloud computing market.[13] The relationship between the decisions to open a data center and invest in an undersea cable are explored in Section 5.

To give a sense of the content of these datasets, detailed summary statistics of the GBRS dataset are presented in Table 1 and a map of data center locations is given in Fig. 2.

Table 1 demonstrates the rapid growth in the number of deployed undersea cables from 2005 to 2021: the cumulative total cable length of the network more than doubled from 572,500 km to 1,301,300 km. This increase was driven by an increase in the number of cables from 180 to

[11] The largest 15 owners (by the number of cables in which the owner has a positive ownership stake) in the TeleGeography dataset are presented in Table A.1.

[12] Some fields, such as construction costs and design capacity are available for about half of the cables in the dataset. Ownership data are widely available, though detailed ownership shares (for multi-firm cables) are unfortunately sparse. In some instances, we are able to add hand-collected cost, length or ownership data.

[13] The count data we use corresponds to the number of "cloud regions." A cloud region is a physical location (e.g., Palo Alto, US) where providers typically cluster multiple data centers (e.g., AWS operates three data centers in the Palo Alto cloud region).

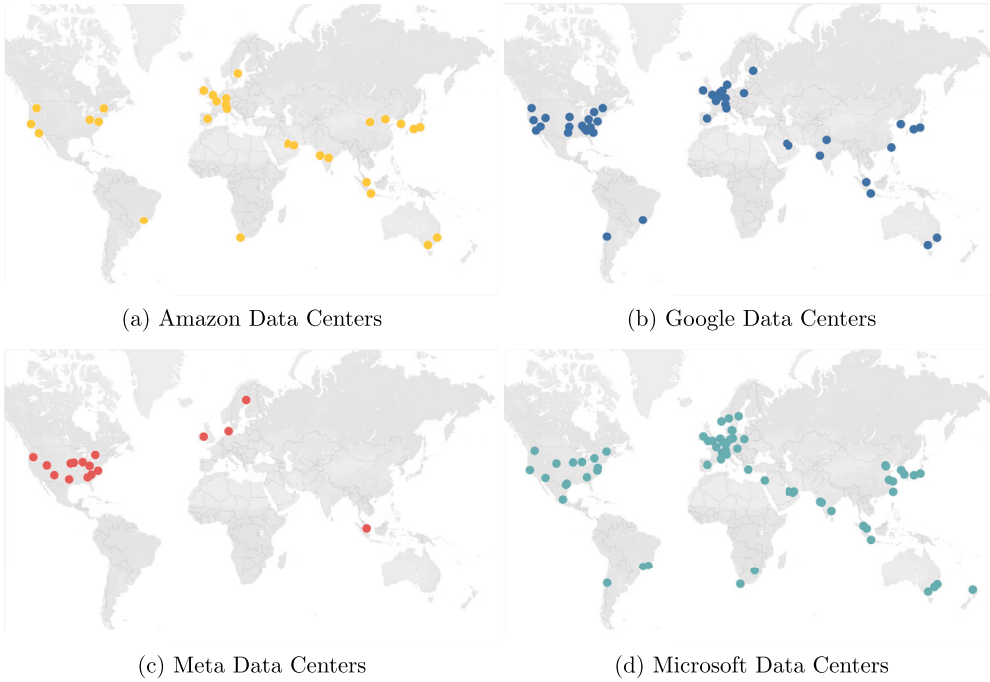


Fig. 2. Map of Datacenter Region Locations as of 2022.

Table 1
Descriptive Statistics on Undersea Cable Supply and Demand.

	New		Cumulative			Bandwidth		New	Cumulative
Year	Cables	'000s km	Cables	'000s km	Cities	Potential	Used	Data Centers	
2005	13	24.9	180	572.5	549	87.1	1.8		
2006	11	31.7	191	604.2	603	89.2	2.6	2	2
2007	11	7.2	202	611.4	623	100.5	4.3	3	5
2008	20	44.3	222	655.7	665	121.0	6.7	1	6
2009	17	69.1	239	724.8	694	142.3	10.2	4	10
2010	14	61.0	253	785.7	726	239.5	15.9	4	14
2011	14	29.4	267	815.1	753	309.8	23.3	7	21
2012	22	61.6	289	876.7	820	470.7	34.1	6	27
2013	11	16.9	300	893.6	834	748.4	49.6	2	29
2014	15	30.2	315	923.9	890	981.2	71.2	10	39
2015	9	19.3	324	943.1	923	1,174.6	99.5	6	45
2016	16	64.7	340	1,007.8	949	1,479.7	145.4	12	57
2017	15	74.1	355	1,081.9	984	1,836.6	224.5	13	70
2018	18	74.0	373	1,155.9	1,040	2,359.3	343.0	12	82
2019	23	31.5	396	1,187.4	1,097	2,521.5	508.5	13	95
2020	21	68.6	417	1,256.0	1,152	3,171.1	715.5	15	110
2021	17	45.3	434	1,301.3	1,189	3,928.3	937.2	8	118

Note: Authors' calculations from TeleGeography's GRBS dataset (see text for details). Columns 2-5 show the number of new cables entering service in a given year, their total length (in thousands of kms), and cumulative totals. Column 6 contains the number of unique cities with an undersea cable landing point; column 7 contains the network's potential capacity, measured in Tbps, while column 8 contains the total used bandwidth (both represent inter-regional figures). Columns 9 and 10 provide counts (net new and cumulative) of data center regions from major content providers contained in TeleGeography's data (this data series begins in 2006).

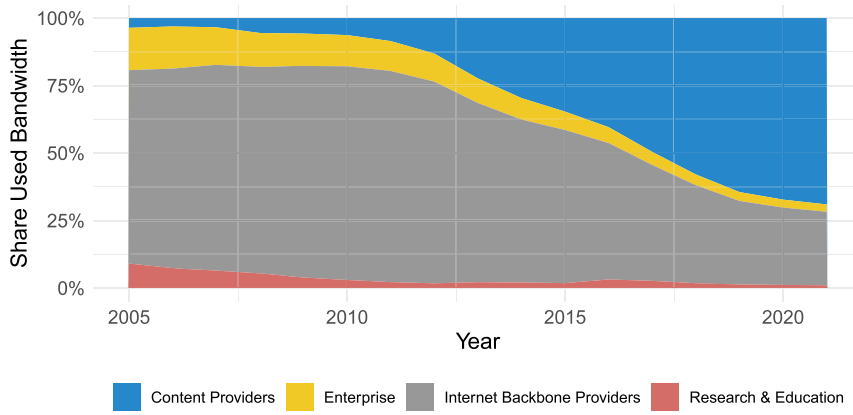
434. The network's potential inter-regional capacity increased by two orders of magnitude over this time frame, from 87.1 to 3,928.3 Tbps.

Table 1 also details some measures of the increase in bandwidth demand over the same time frame. The number of distinct metropolitan areas directly connected to an undersea cable increased from 549 to 1,189, and the inter-regional bandwidth flowing over undersea cables increased from 1.8 to 937.2 Tbps. Such an increase is evident both globally, as well as across all measurable sub-geographies. In addition, it was accompanied by an increase in the number of data center regions operated by major content providers, from 2 in 2006 to 118 in 2021. As we will see in Section 3, much of the increase in bandwidth was driven by the same firms that were building data centers.

Auxiliary data

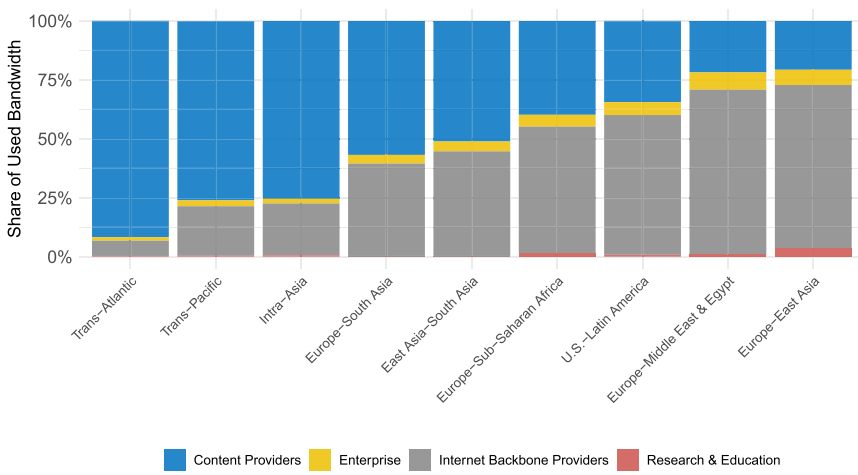
For the analysis in Section 5, we supplement TeleGeography's GBRS dataset with a few auxiliary sources. These include several determinants of data center location or cable landing location decisions at the country-year level.

From the CEPII Gravity Database, a repository of macroeconomic and trade-related variables that are suitable for the estimation of the



Notes: This figure depicts the share of global international bandwidth by buyer type and over time.

Fig. 3. Global Bandwidth Shares. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)



Notes: This figure depicts the share of international bandwidth by buyer type in 2021, broken down by subsea route. Authors' calculations from TeleGeography data and reports.

Fig. 4. Route Bandwidth Shares in 2021.

determinants of international trade (Conte et al., 2021), we source data on country characteristics, such as annual GDP figures, WTO membership, and EU membership. We complement this data source with a panel on corporate tax rates at the country level from the Tax Foundation.[14]

As data centers are energy-intensive facilities, we source from Our World in Data (ourworldindata.org) a country-level panel of electricity costs. With many data center developers and operators increasingly demonstrating concern about the emission intensity of their industry, we also use a country-level panel of the fossil-share of electricity generation from the same source.

3. Content providers' bandwidth demand

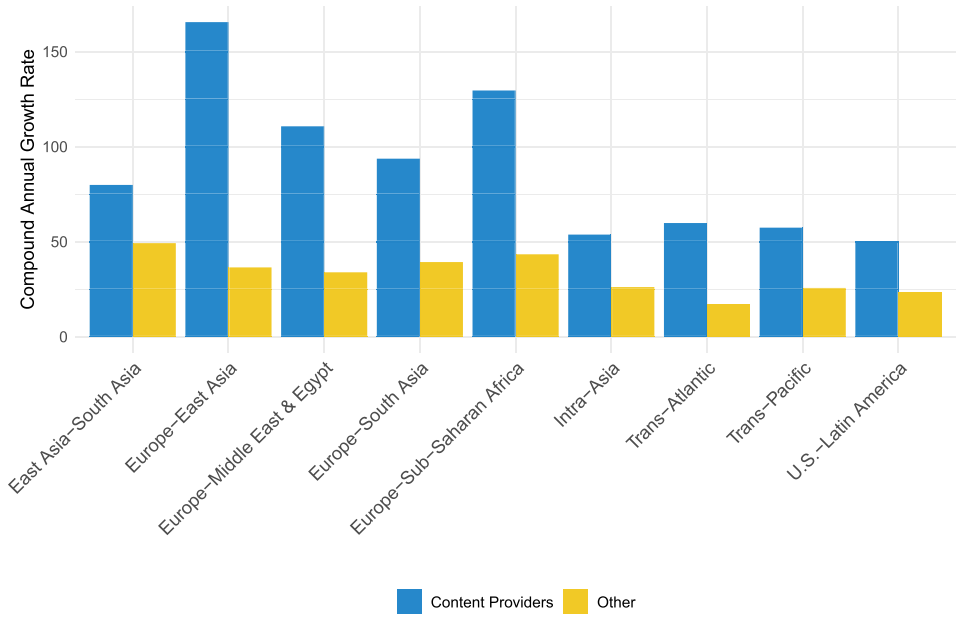
This section details the growing role played by content providers as buyers in the market for international bandwidth over the period 2005-2021.

In the previous section, we pointed out that the growth in international bandwidth demand over our sample period has been tremendous. It turns out that much of this growth has been driven by growth in demand from content providers, as can be seen in Fig. 3. Fig. 3 depicts the changing shares of international bandwidth demand over the sample period. The major trend is clear: beginning around 2010, the share of international bandwidth demand attributed to content providers began increasing from a level of roughly 5% to nearly 70% by 2021. Globally, this growth in demand share attributable to content providers has come primarily at the expense of internet backbone providers' demand share, which decreased from roughly 60% in 2005 to 25% in 2021. The demand shares of enterprise and research and education network customers have also diminished, but from much lower starting points.

At the level of the nine distinct major cable subsea routes, there is a higher degree of heterogeneity. Demand shares for international bandwidth in 2021 by route are visible in Fig. 4. The outsized global role of content providers is explained by their predominance over the three largest routes: Trans-Atlantic, Trans-Pacific, and Intra-Asia. On these routes, the demand share of content providers is in the 75-90% range. However, the demand share of content providers is significantly lower on other routes, particularly Europe-East Asia, Europe-Middle East & Egypt, and U.S.-Latin America.

An examination of demand growth rates by buyer-type and route indicates that the heterogeneity shown in Fig. 4 may be relatively short-lived. We calculate the compound annual growth rate in bandwidth demand from 2016 to 2021 separately for content providers and all other buyer types and depict them in Fig. 5. Across every route, the rate of demand growth from content providers significantly outpaces that of all other providers combined. In many cases, this difference in compound annual growth rate is by a factor of 4 or 5. A simple forecast of bandwidth shares that accounts for this difference in growth

[14] https://taxfoundation.org.



Notes: Depicted are the compound annual growth rates in bandwidth demand from 2016-2021 by subsea route, calculated separately for Content Providers, and all other types of purchaser. Authors' calculations from TeleGeography data and reports.

Fig. 5. Bandwidth Growth Rates, 2016 – 2021.

rates indicates that content providers will make up the largest share of bandwidth demand on every route in the near future. Further details on growth rates by period, buyer type, and route are provided in Appendix Table B.2.

Taken together, the figures and analysis in this section unequivocally demonstrate both the increased importance of content providers as users in the market for international bandwidth in recent years, and that their outsized importance is likely to further increase in coming years. In the next section, we analyze the changing role of content providers on the supply side of the market: cable investment and ownership.

4. The deployment of Internet infrastructure

In this section, we examine the changing role that content providers have played on the other side of the market for international bandwidth: the increasing propensity for content providers to vertically integrate upstream into owning and operating their own cables. We begin by discussing the types of ownership and investment available to content providers, and describe investments (historical and planned) to date. Next, we use available data to construct traditional measures of historical ownership concentration in cables, before analyzing trends in ownership concentration over the years of our data. Finally, we examine recent (and projected) cable investment trends by content providers and other types of cable owners.

A noteworthy finding of our analysis is the evolving landscape of cable investments. While content providers have been important users of international bandwidth for nearly a decade, they have not historically been investors in new cables, particularly in terms of capital expenditure for new construction. However, this landscape of investment in new cables has sheen a dramatic shift in recent years. Content providers are building private cables, are increasingly joining consortia to build cables, and are committing to up-front capacity purchase agreements that support new construction. When these different channels are taken together, content providers account for a significant share of capital investment for recent cables and those planned in the immediate future; on the major Trans-Atlantic and Trans-Pacific routes, the majority of very recent and planned cables are either solely-owned or invested in by content providers.

4.1. Ownership

We begin by noting that undersea fiber optic cables represent major infrastructure investments. The cost to build a cable varies with cable length and capacity (i.e. the amount of bandwidth, in Gbps, that can theoretically carried by the cable), with total construction costs typically in the hundreds of millions of dollars. Most cables are sole-owned by a single institution, typically a private firm but occasionally a government; due to the high costs however, many of the largest cables are constructed and owned by a consortium of investors. These investors are usually private firms, but may include governments or other non-firm institutions. Investment in cables can also take place indirectly, in the form of an up-front commitment to purchase a particular amount of capacity (with prices and time frame specified by contract). This dynamic has parallels in other major infrastructure markets: e.g. the widespread use of power purchase agreements in the market for electricity generation; or the advanced purchase orders between the US Government and various COVID-19 vaccine developers in the context of Operation Warp Speed. As in these other cases, a contractual commitment on the part of a content provider to purchase bandwidth significantly reduces the financial uncertainty around building a cable, and is a significant aid to investment in new cables.

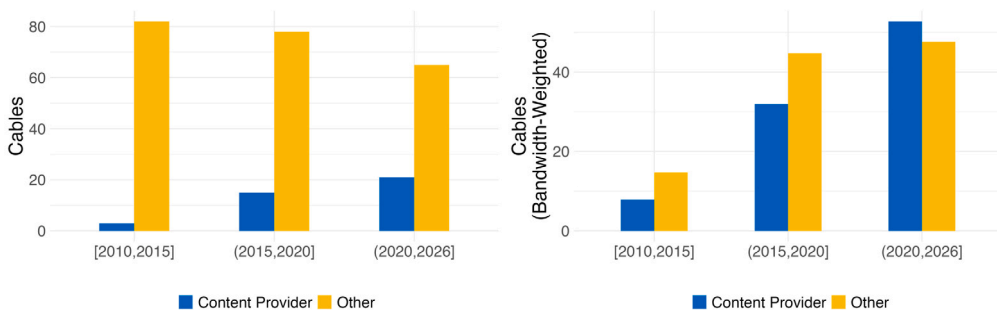
In our data, content providers make all three types of investments in cables: as members of consortia, as sole owners, and as major capacity customers. Details on content provider ownership stakes are shown in Table 2: the four major content providers have 50 ownership stakes across 44 cables. The preponderance of these ownership stakes are as consortium members, though Google solely-owns eight cables and Meta one. While all four major content providers have cable ownership stakes, Microsoft and Amazon have significantly fewer than Google and Meta. This difference may represent differences in ultimate demand (e.g. content delivery vs. cloud services), differences in geographic reach, and/or differences in corporate strategy (e.g. differing appetite for vertical integration vs. buying bandwidth from other providers).[15]

[15] Bandwidth requirements from search and social media far exceed those of cloud services.

Table 2
Content Provider Investment.

		Investment Type			Investment Date		
Content Provider	Total	Consortium Member	Purchase Agreement	Sole Owner	Earliest Year	Mean Year	Latest Year
Amazon	5	2	3	0	2018	2019.8	2023
Google	22	14	0	8	2010	2019.8	2024
Meta	17	15	1	1	2016	2021.2	2024
Microsoft	6	4	2	0	2015	2018.0	2023
Total	50	35	6	9	2010	2020.1	2024

Notes: This table details the number of investment stakes in (existing or under construction) undersea cables by Amazon, Alphabet/Google, Meta, and Microsoft. Column 1 contains the total, while columns 2-4 present subtotals by differing types of investment stakes. Columns 5-7 present statistics on the date of investment decisions. The data are at the ownership stake (cable-owner) level, and represent 50 ownership stakes in 44 cables. Authors' calculations from TeleGeography data and reports; data from 2022 onward is based on projected cable completions.



(a) Cables (b) Bandwidth-Weighted Cables

Notes: The left panel plots the number of new cables entering into service over time, by the status of whether the cable has an investment by a content provider. The right panel repeats the exercise, but weights each cable by the bandwidth on that route in 2021. Authors' calculations from TeleGeography data and reports; data from 2022 onward is TeleGeography's projected cable completions based on under-construction and announced cables.

Fig. 6. Investment by Content Providers.

Finally, we point out that the increase in investment by content providers is a recent phenomenon, matching the recent increase in bandwidth demand by the same content providers discussed in Section 3. The first content provider cable investment in our data took place in 2010, with Google as a consortium member in the Unity cable between the U.S. and Japan. However, it is only in 2018 and later years that 45 of the 50 cable investments detailed in Table 2 took place.

4.2. Investment in undersea cables

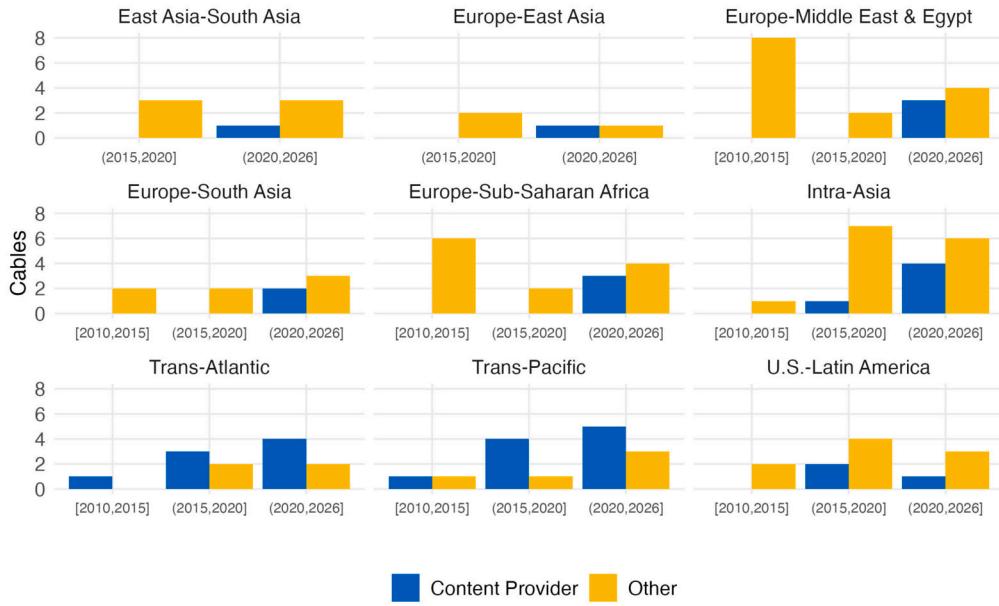
Motivated by the findings presented in Table 2, we now more closely examine the trends in cable investment by content providers as opposed to other cable owners/investors. In particular, we will show how the number of new cables featuring content provider investment has changed over time. First, we will consider the total number of cables; second, an alternative measure that will weight cable investment by the bandwidth demand on each route in an attempt to tease out the relative size of the markets in which content providers are investing; third, a breakdown in the absolute number of cables over time by the route of investment. This section uses the ownership data used in Section 4.1 combined with the universe of cable profiles available from the GBRS data. Data for the years 2022-2026 are based on TeleGeography's projections of entry by announced and under-construction cables as of year-end 2021.[16]

The left panel of Fig. 6 plots the number of cables built across three discrete periods by whether or not those cables received *any investment* from a content provider. For this figure, we consider a content provider to have invested in a cable if any of the three investment forms discussed above are present. As an example, between 2016 and 2020, 78 cables entered service without any content provider investment, and 15 cables with content provider investment entered service. Two patterns are immediately discernible from the left panel of Fig. 6: first, the number of cables being built has continued to hold steady over our sample time frame, consistent with the patterns discussed in Section 2; second, the share of new cables with content provider investment has been steadily rising. To be precise, the share of new cables with a content provider investment has risen from 3.5% of new cables from 2010 through 2015, to 24.4% of new and projected cables from 2021 through 2026.

The right hand of panel of Fig. 6 features a similar plot, but weights the number of cables built on each route by the 2021 bandwidth demand on that route. That is, routes with above-average bandwidth demand are upweighted and routes with below-average demand are downweighted. The contrast between the left- and right-hand panels of the figure emphasizes that content provider investment in cables is disproportionately focused on high-bandwidth demand markets.

This tendency can also be seen in Fig. 7, which also plots unweighted cables but breaks down the results by route. It can be seen from this figure that the trend of increased content provider investment holds on eight of the nine routes. Indeed, the trend is quite pronounced both on the three routes with the highest share of content provider demand (Trans-Atlantic, Trans-Pacific, and Intra-Asia, see Fig. 4), and also on

[16] More details on the data used are available in Section 2.2.



Notes: This figure plots the number of new cables entering into service over time by route, and also by the status of whether the cable has an investment by a content provider. Authors' calculations from TeleGeography data and reports; data from 2022 onward is TeleGeography's projected cable completions based on under-construction and announced cables.

Fig. 7. Investment by Content Providers and Route.

many routes where the share of content provider demand is lower and that received less investment in general in earlier periods. These trends re-emphasize the growing role that content providers are playing and are projected to play across all routes as investors and owners. Together with the demand growth rates plotted in Fig. 5, it is clear that content providers, and in particular Amazon, Alphabet/Google, Meta and Microsoft, are playing an increasingly prominent role on both the supply and demand sides of the international bandwidth market.

These trends in investment combined with the small number of content providers suggest *prima facie* that market power from concentrated ownership could become an issue in this market in the coming years, an issue considered in some detail in Appendix C. In the next section, we turn to an empirical investigation of one of the major factors that has been reported to be responsible for the rising trend of content provider investment in cables: an attempt to connect their proprietary data centers more directly than allowed by the prior cable network.

5. The role of data center infrastructure

This section explores factors that drive content providers' demand for capacity at the global level and their incentive to vertically integrate, with a focus on the role played by data centers. We first show that the growth in content providers' bandwidth demand is associated with the global roll-out of data centers: in particular, inter-data center traffic is an important driver of aggregate data traffic. Next, we find evidence that content providers invest in internet cable infrastructure in a selective manner that connects their network of data centers. Data centers can be deployed in locations that may not be well connected to the existing "public" internet cable infrastructure because of their smaller size (in terms of bandwidth traffic) or other cost consideration (e.g., corporate tax regimes).

Inter-Data Center Demand. Content providers have high demand for bandwidth between their proprietary data centers. To investigate how data enter investment decisions affect aggregate traffic, we explore within-market variation in data center investment using an event study framework. For a given market, which we define as a country pair, we analyze how changes in the number of data centers operated by content providers affect data traffic in the market.[17]

The unit of analysis is the market-year. As the effects we estimate are likely persistent and heterogeneous over time, we consider a dynamic (event study) specification. To highlight the role of inter-data center traffic, we consider two possible event treatments: (i) if at least one content provider operates a positive number of data centers in *both* countries comprising the market, or (ii) if at least one content provider operates a positive number of data centers in *only one* country in the market. Let Q_{mt} denote the total used bandwidth in market m at time t.[18] We consider the following dynamic specification

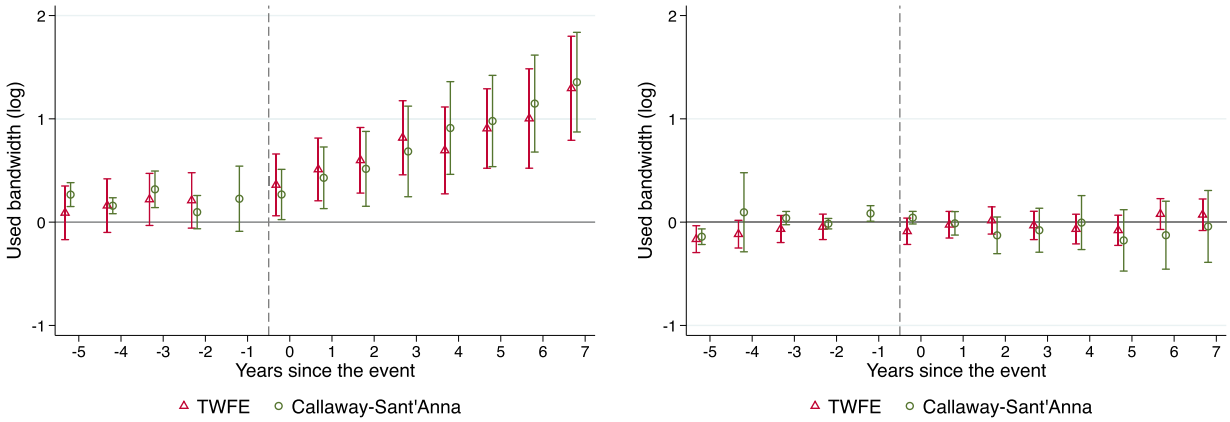
$$\log(Q_{mt}) = \sum_{\tau=-5}^{7} \delta_\tau DC_{m,t-\tau} + \gamma_m + \eta_t + \epsilon_{mt} \tag{1}$$

where $DC_{m,t-\tau}$ is an indicator variable that equals 1 if market m at time t was treated τ periods ago for $\tau > 0$ (or respectively, will be treated τ periods in the future, if $\tau < 0$). γ_m and η_t are market and time fixed effects.

This specification allows for both dynamic effects, such as a delay in the effect of the treatment, and for detecting the presence of pre-trends in bandwidth growth prior to the treatment. The coefficients after the treatment has occurred (δ_τ for $\tau \geq 0$) capture the dynamic effects of data center presence on used bandwidth. The terms δ_τ before the event has occurred (for $\tau < 0$) provide a placebo test. In the absence of anticipation effects (i.e., content providers building data centers in markets that are anticipated to grow), model misspecification, or omitted confounding variables, these pre-event terms should not have a trend in τ. When implementing this specification, the omitted category is $\tau = -1$ so that all cumulative effects are relative to the year before the treatment. We

[17] The decision to define a market as a country-to-country pair reflects data limitations and follows the approach in Caoui and Steck (2024).

[18] In principle, if all content providers ceased to operate data centers in the relevant market, a market might move from treated to untreated status. In practice, as this industry is still in its nascent growth phase, we do not observe any data center exits.



(a) Data center presence in both countries

(b) Data center presence in a single country

Fig. 8. The effects of data center presence on total used bandwidth at the country-pair level. The left panel shows the effect of a content provider operating data centers in both countries in the pair on used bandwidth between the two countries. The right panel shows the effect of a content provider operating data centers in a single country in the pair on used bandwidth between the two countries. Results are from an event study analysis using a TWFE and a heterogeneity-robust estimator proposed by Callaway and Sant'Anna (2021). Country-pair and year fixed effects are included. Standard errors are clustered at the country-pair level. Confidence bands show the uniform sup-t confidence intervals adjusted for multiple hypothesis testing.

set $\tau \in \{-5, \ldots, 7\}$ to consider dynamic effects up to 5 years before the treatment and up to 7 years after the treatment.

Fig. 8 shows the event study plots, using a two-way fixed effect as well as a heterogeneity-robust estimator (Callaway and Sant'Anna (2021)). The left panel corresponds to treatment (i), whereas the right panel to treatment (ii). We find that, if a content provider operates data centers at both ends of a market, used bandwidth increases substantially in the years following: e.g., by 170% within 5 years relative to untreated markets. In the periods prior to the event, there is no statistically significant difference in the trend of used bandwidth between treated and control markets. On the other hand, we do not find significant changes in bandwidth growth when we examine treatment (ii) (the right panel of Fig. 8). If a content provider operates a data center in only one end of the country-country market, there are no effects on bandwidth demand growth.

Cable Investment and Data Center Presence. We next examine the factors that explain content providers' recent strategy of upstream vertical integration into undersea cable provision and operation.[19] Vertical integration is likely driven in part by an effort on the part of content providers to avoid holdup by traditional cable owners, especially on new and less competitive routes. Other possible explanations include large investment needs at the largest firms not being met by the market; an attempt to increase bargaining power on the wholesale bandwidth market; a desire to exclude rivals from key investments or routes; and a benefit from tighter coordination across content provision and transportation.

Here, we focus on the role played by data center locations. In particular, one possible motivation for content providers' growing investment in undersea cables is to connect their networks of data centers. They can do so more efficiently than by relying on traditional backbone providers, especially if data centers are in locations without existing public undersea cable landings and a risk of hold-up is present.

Historically, undersea cables have been sited in order to minimize the costs of communicating between two disparate terrestrial networks; their landing points were determined by the location of demand for that communication and by coastal geography. In contrast, data centers serve different needs and have different cost structures (e.g., electricity costs), making the optimal placement of data centers in many cases fairly distant from undersea cable landing points. By making direct investments in the construction of new cables, content providers can shape network topographies to align with their specific network needs.

Anecdotal evidence supports the hypothesis as being a key motivation for content providers' cable investment. For instance, content provider cables such as AEC-1 and EXA Express have landing points in Ireland, a relatively small bandwidth market, but one where many data centers are situated. Similarly, Google's Curie cable establishes a direct link between the United States and the company's data center in Chile. Content providers also express a preference for developing new cables with configurations that facilitate direct connections between primary landing points while bypassing intermediary locations. This approach serves to minimize latency and equipment expenses. An instance is the FASTER trans-Pacific cable, specifically tailored to fulfill the requirements of Google, the exclusive proprietor of the segment connecting Taiwan to the United States. Google's operations include a data center in Taiwan, and the company possesses an express fiber pair on the FASTER cable, reinforcing the direct linkage between Taiwan and the United States.

In order to empirically test this hypothesis, we compare the characteristics of the cable landing points chosen by content providers to those chosen by traditional cable owners (backbone providers). In this exercise, we again focus on the presence of content provider data centers in determining the location choice. We model landing decisions using a multinomial logit specification as follows. Let r denote regions (e.g., North America, Europe, Middle East), c denote countries, t denote time, and $Land_{icrt}$ be a dummy variable for whether cable i lands in country c of region r (at time t).[20] Conditional on cable i landing in region r, the probability that the cable lands in country c is given by

$$\Pr\left(Land_{icrt} = 1\right) = \frac{\exp\left(\beta_1' \mathbf{X}_{ct} + \beta_2 n_{ct}^{DC} \times CP_i\right)}{\sum_{c' \in C(r)} \exp\left(\beta_1' \mathbf{X}_{c't} + \beta_2 n_{c't}^{DC} \times CP_i\right)} \tag{2}$$

[19] It is worth noting that there are historical examples of vertical integration between content providers and telecommunications firms (e.g., NBC and Comcast).

[20] For a given cable, the entry date is fixed, therefore the time subscript is redundant. We nonetheless use it to clarify that the variables $\mathbf{X}_{ct}$ and n_{ct}^{DC} refers to country characteristics and the number of data centers in country c by time t.

Table 3
Cable Landings and Data Center Presence.

	Dependent Variable: *Landing_ij*			
	(1)	(2)	(3)	(4)
GDP (log)	0.246***	0.246***	0.124*	0.136*
	(0.0336)	(0.0340)	(0.0523)	(0.0534)
EU member	-0.849***	-0.209	-0.305	-0.306
	(0.110)	(0.209)	(0.219)	(0.212)
WTO member	0.128	0.451*	0.391*	0.423*
	(0.181)	(0.189)	(0.196)	(0.209)
Used bandwidth (log)	-0.00852	0.0188	0.139**	0.143**
	(0.0268)	(0.0276)	(0.0515)	(0.0528)
Growth in used bandwidth (log)	0.0967*	0.104*	0.118*	0.114*
	(0.0399)	(0.0482)	(0.0569)	(0.0544)
Number of data centers ×				
Non-CP Cable	0.0637***	0.0360*	0.0554**	0.00542
	(0.0148)	(0.0171)	(0.0214)	(0.0139)
CP Cable	0.105***	0.0778**	0.0949***	0.0750*
	(0.0243)	(0.0254)	(0.0270)	(0.0341)
CP FE	Yes	Yes	Yes	No
Region FE	No	Yes	No	No
Region × Year FE	No	No	Yes	No
Cable × Region FE	No	No	No	Yes
Observations	8535	8535	8500	8221

Note: The unit of observation is a cable-country pair. Standard errors are clustered at the cable level. "CP" stands for content providers.

where $C(r)$ is the set of countries in region r, $\mathbf{X}_{ct}$ is a vector of time-varying country characteristics (GDP, data traffic growth, etc), n_{ct}^{DC} is the number of data centers located in country c by time t, CP_i is a dummy for whether cable i is owned by content providers.[21]

The results are shown in Table 3. Columns (1) to (3) include various levels of fixed effects: by content provider identity (i.e., Amazon, Google, Meta, or Microsoft), by region, by region and time. Column (4) includes a cable-region (equivalently content provider-region-time) fixed effect: i.e., we identify the effects of $\mathbf{X}_{ct}$ and n_{ct}^{DC} from variation across countries in a given region and time period. We find that internet cables are more likely to land in countries with higher income and less likely to land in non-WTO countries. Countries with higher demand (measured by the level and growth rate of used bandwidth traffic through the country) are also more likely to host a cable landing.

Finally, we find that cables owned by content providers are more likely to land in countries where these content providers operate data centers. This relationship does not hold for cables not owned by content providers. The positive correlation between content providers' cable landing locations and the presence of data centers indicates that their decisions around cable ownership and investment are intertwined with their global networks of data centers. In particular, content providers evidently prefer to connect locations where they have a data center presence.

Data Center Locations. Next, we examine potential explanations for why content providers choose to connect their data center infrastructure via private undersea cables rather than relying on the existing public backbone network. To do so, we investigate the factors driving data center location choices. Do these locations differ from the typical location where public cable infrastructure lands?

Following a similar specification as in the previous analysis, we compare the characteristics of the locations where data centers are built. Data center location choices are modelled as a multinomial logit model. Let DC_{icrt} denote a dummy variable for whether content provider i deploys a data center in country c in region r at time t. Conditional on a data center being deployed in region r at time t, we model the probability that it is built in country c as

[21] The set $C(r)$ is restricted to countries where at least one cable has ever landed by 2021.

Table 4
Data Center Location Choice.

	Dependent Variable: DC_ij			
	(1)	(2)	(3)	(4)
GDP (log)	0.713***	0.763***	0.588**	0.525**
	(0.132)	(0.146)	(0.200)	(0.178)
EU member	-0.00126	2.636*	3.039*	2.767*
	(0.362)	(1.192)	(1.465)	(1.348)
Used bandwidth (log)	0.00189	0.133	0.304	0.335
	(0.0882)	(0.0942)	(0.193)	(0.175)
Growth in used bandwidth (log)	0.254	0.267	0.269	0.230
	(0.156)	(0.150)	(0.160)	(0.129)
Number of data centers	0.222*	0.138	0.176	0.0982
	(0.0989)	(0.0820)	(0.111)	(0.0831)
Number cables landing	0.00679	0.0187	0.0250	0.0177
	(0.00938)	(0.0109)	(0.0134)	(0.0115)
Number CP cables landing	-0.176	-0.233	-0.237	-0.224
	(0.135)	(0.143)	(0.168)	(0.143)
Electricity price (log)	0.122	-0.0234	-0.356	-0.208
	(0.205)	(0.282)	(0.505)	(0.456)
Fossil share in electricity mix	-0.00437	0.00629	0.0144	0.00710
	(0.00792)	(0.00893)	(0.0150)	(0.0134)
Corporate tax rate	-0.0283	-0.0636*	-0.0623*	-0.0591*
	(0.0253)	(0.0280)	(0.0306)	(0.0258)
Latitude (absolute value)	-0.0193*	-0.00564	-0.00370	-0.000182
	(0.00903)	(0.0149)	(0.0154)	(0.0134)
CP FE	Yes	Yes	Yes	No
Region FE	No	Yes	No	No
Region × Year FE	No	No	Yes	No
CP × Region × Year FE	No	No	No	Yes
Observations	1395	1395	1383	1321

Note: The unit of observation is a content provider-year-country. Standard errors are clustered at the content provider-year-regional level. "CP" stands for content providers. No data centers are built in WTO non-member countries, we do not include this indicator.

$$\Pr\left(DC_{icrt}=1\right)=\frac{\exp\left(\alpha'\mathbf{X}_{ct}\right)}{\sum_{c'\in C(r)}\exp\left(\mathbf{X}_{c't}\right)}, \tag{3}$$

where $C(r)$ is the set of countries in region r, $\mathbf{X}_{ct}$ is a vector of time-varying country characteristics: these include measure of economic activity such as GDP and used bandwidth (at the country level over time), regulation (EU membership), digital connectedness (proxied by the number of cables landing in the country), and data center cost factors (electricity price, fossil fuel share in electricity mix, corporate tax rate, and latitude as a proxy for average temperature). Regression results estimating Equation (3) are presented in Table 4.

We find that data centers tend to be located in countries with higher income, as measured by log of GDP. Interestingly, these locations do not necessarily have high data traffic levels: e.g., the level and growth of used bandwidth is not a significant predictor of data center entry. This is arguably consistent with the absence of pre-trend found in Fig. 8. This fact also indicates that content providers locate data centers in "isolated" countries with smaller data traffic footprints that may not be as well-connected to the network of public cables. In Europe, we find that EU membership is a strong predictor of data center entry.[22] This effect may be explained by several factors related to the economic, legal, and operational environment that EU membership provides: e.g., regulatory harmonization, where harmonization reduces the complexity and costs associated with complying with different data protection and privacy regulations, and EU-level initiatives to promote digital economy growth, including funding or policy support for cloud computing and data infrastructure. We find that favorable tax regimes are asso-

[22] This coefficient is identified from cross-sectional variation in EU membership across countries (e.g., from non-members such as the UK, Norway, Iceland, Albania) and time-series variation in membership (Croatia joined the EU in 2013).

ciated with greater likelihood of data center entry, but neither electricity costs nor the use of renewable sources in electricity generation are statistically significantly associated with data center entry in our data.

To provide intuition for the size of these effects, we compute marginal effects. The average probability of data center entry in country c of region r at time t is 8.7%. If GDP is doubled, the average probability of data center entry increases to 10.7% (or a 23% increase). Moving from the 25th percentile of corporate tax rates (18%) to the 75th percentile (30%) reduces the average probability of data center entry to 4.9% (or a 42% decrease). Within-Europe, EU membership has a significant effect on data center entry, raising the average probability of entry to 26%. We note that the large effect of EU membership might capture unobserved factors that are correlated with EU membership.

6. Implications and discussion

The previous sections illustrate the growing role content providers play as consumers of bandwidth capacity and how an increasingly large share of international internet traffic is carried over their private networks. In this section, we discuss some potential medium and long-term implications of these trends. Our discussion focuses on how these trends might impact quality of service in bandwidth provision, innovation and investment in internet infrastructure, the adoption of next-generation technology standards, and the resilience of internet networks.

Latency and Quality of Service. The involvement of content providers in the deployment of internet infrastructure may result in significant improvements in service quality. Because content is increasingly stored in data centers and content delivery networks (CDNs) closer to consumers (geographically and in terms of network topology), latency should be reduced and user experience improved.[23]

Prior to the rise of content providers' private networks of cables, internet requests from downstream users were carried through the public internet. This traffic passed through various independent nodes and traversed potentially long distances in terms of geography and/or network topology. While the open internet architecture still exists today, the public infrastructure carries traffic over shorter physical distances. Most of the content accessed by users today is primarily stored in nearby data centers and content delivery networks. Internet requests briefly pass through the public network managed by the ISP, before reaching on-ramps designed to direct traffic to and from a content provider's data centers and CDNs. As a result, many requests no longer need to navigate the vast expanse of the open public internet and data increasingly only spends a short amount of time on the public internet network.

Some content providers, especially video-streaming firms, have strategically embedded content servers directly within ISP networks. This placement ensures rapid access for broadband subscribers, further reducing latency and enhancing the overall quality of service. In this evolving landscape, content providers' involvement in infrastructure deployment likely leads on the whole to more efficient and lower-latency information delivery networks.

Innovation and Traffic Transparency. However, the rising concentration of traffic in private networks owned by content providers may adversely impact innovation and investment in the public internet infrastructure.

First, as major content providers gain influence in the bandwidth market, either as wholesale buyers or through vertical integration, a dynamic akin to the "preference externalities" described in Waldfogel (2003) and George and Waldfogel (2003) can emerge. In this situation, the concerns for the social planner are that investment in internet infrastructure would cater too specifically to the preferences of major content providers. In the most extreme case, this targeted investment might entrench the market dominance of large content providers, disadvantaging current and potential competitors, and ultimately harming consumers. Additionally, as content providers invest in their own private cable networks, they remove a significant fraction of demand from public internet infrastructure, potentially making it uneconomical due to high fixed costs. This shift could lead to reduced investment and slower bandwidth growth for other types of users relying on public internet infrastructure.[24]

Relatedly, as more data traffic is carried over private networks, public internet providers lose visibility over an increasingly significant portion of global internet traffic. This development raises concerns because internet providers may find themselves lacking the information needed to efficiently enhance internet services and sustain the development and operation of their cable networks. The diminished visibility of internet traffic to public internet providers could potentially solidify the market position of dominant content providers, altering the dynamics of the internet infrastructure landscape.[25]

As content providers handle larger swaths of internet traffic and private networks continue to expand, the balance between innovation, traffic transparency, and profitability in internet infrastructure may experience significant transformation, raising crucial questions for the future of investment in public internet infrastructure.

Technology Standards. The growing role played by content providers may also impact the speed of adoption of open technology standards.

On the one hand, the rise of a small number of privately operated global networks may reduce incentives for the development and adoption of better technology standards. One notable example is the reduced incentive for carriers to transition from IPv4 to IPv6 (OECD (2014)).[26] Content providers rely heavily on Network Address Translation (NAT) to manage their private networks using IPv4. NAT allows multiple devices within a private network to share a single public IPv4 address. Because these companies use NAT to share a limited number of public IPv4 addresses among many devices on their private networks, the scarcity of IPv4 addresses has become a less pressing concern. As a result, the incentive for content providers to adopt IPv6, which was introduced to solve the problem of IPv4 address exhaustion, has been significantly attenuated. Industry experts have noted that the widespread adoption of IPv6 may have been delayed partly because content providers have found ways to cope with the limited availability of IPv4 addresses using NAT in their private networks; this slowing of the global adoption of IPv6 has imposed a negative externality on other participants in the internet supply chain (Register (2022)).

On the other hand, content providers may have the capacity to implement technological changes that suit their networks and applications without waiting for global standard development, which can be a time-consuming process of collaborative development and dissemination. Content providers' greater degree of control over their private networks imparts the ability to promptly address some specific needs outside of the open internet protocol development channels; this takes shape as a greater agility in network development. While their pro-

[23] For a definition of CDNs, see footnote 2.

[24] Analyzing potential negative spillovers of content provider's vertical integration on other types of bandwidth users would be an important avenue for future research. Additional data on latency or quality of service at the cable level would help tackle this question.

[25] One practical example of this shift can be seen in the case of Apple's Private Relay. While it is seen by customers as enhancing privacy and security, some carriers have argued that it simultaneously erodes their ability to effectively invest in network improvements (Register (2022)).

[26] IPv4 uses 32-bit addresses, which allows for approximately 4.3 billion unique IP addresses. Due to the increasing number of devices connected to the internet, IPv4 addresses have become scarce. IPv6 uses 128-bit addresses, providing an astronomically larger number of unique addresses (approximately 340 undecillion addresses). This abundance of addresses ensures that every device can have a unique IPv6 address.

prietary technologies must maintain interoperability with the broader internet, this trend toward increased centralization blurs the boundaries between the network and application layers, challenging the traditional layering principle in internet architecture.[27] The growing influence of content providers in shaping the adoption of technology standards thus raises complex questions with implications on both the pace and the direction of standardization efforts on the network of public internet infrastructure.

Reduced Decentralization and Network Resilience. The expansion of content providers' private networks also has an impact on overall network resilience. One concern is the reduced decentralization of internet infrastructure, leading to fewer redundancies in the network, and a situation where a smaller number of issues might lead to larger network failures. An increasing portion of global traffic is being carried between data centers and to CDNs over a few global private networks. Real-world examples illustrate the risks associated with a network architecture heavily dependent on a small number of private networks. High-impact outages, such as the AWS outage in 2020, Fastly outage in 2021, Akamai's subsequent outage, and Meta's outage in 2024, showcased how reliance on a single CDN or infrastructure provider can lead to global unavailability of internet services (The Guardian (2021), Register (2021)). These incidents underline the vulnerability introduced by centralization, where a flaw in a distributed algorithm or a failure in a single platform can create a critical point of failure with far-reaching consequences.

7. Conclusion

This paper considers the growing role of large technology firms in the market for upstream internet infrastructure. We document how the rise of large technology platforms in consumer and product markets has been accompanied, in recent years, by their rise as large investors in and proprietors of undersea fiber optic cables.

We leverage novel data on bandwidth usage and investment in internet infrastructure (undersea cables and data center) to establish three stylized facts. First, content providers are very large buyers in global bandwidth markets, surpassing the traditional players in the market, internet backbone providers. Second, they account for an increasingly important share of investment in undersea internet cables. Third, their demand growth and vertical integration into cable operations are tightly linked to their data center operations. In particular, content providers' direct investment in cables allows them to exert an influence on network topology in order to better connect their data centers.

Finally, we discuss potential implications of these trends for the medium and long-term. Content providers' deployment of infrastructure may be accompanied by substantial efficiency gains; but in the longer-run, this pattern of vertical integration may also reduce traffic transparency and incentives for investment in public internet infrastructure. The speed of adoption of new technology standards as well as the resilience of the network could be affected by the fragmentation of the internet into a small number of private networks. While this final discussion is admittedly speculative, we view this paper's main contribution as using data to shine light on recent notable trends in the use of, investment in, and deployment of fundamental internet infrastructure. While the role and conduct of the largest technology firms in downstream markets have received attention in policy discussions in recent years, and in some instances even scrutiny from competition authorities, to date little attention has been paid to trends involving these firms as important players in upstream internet markets.

[27] The layering principle in internet architecture is a fundamental concept that structures the design of computer networks and the internet (e.g., TCP/IP or OSI models). It involves dividing the complex task of network communication into distinct, modular layers, each with its own specific functions and responsibilities. The primary goal of this approach is to promote modularity, ease of maintenance, scalability, and interoperability between different network components and technologies. The network layer is responsible for routing packets of data between different networks and subnets. It determines the optimal path for data to travel from the source to the destination. The application layer is where end-user applications and network services interact with the network. It is used by end-user software such as web browsers and email clients. It provides protocols (e.g., HTTP, FTP) that allow software to send and receive information and present meaningful data to users.

CRediT authorship contribution statement

El Hadi Caoui: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Andrew Steck:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Andrew Steck reports financial support was provided by NET Institute. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Additional data details

This appendix contains additional details on some aspects of the market context for the interested reader.

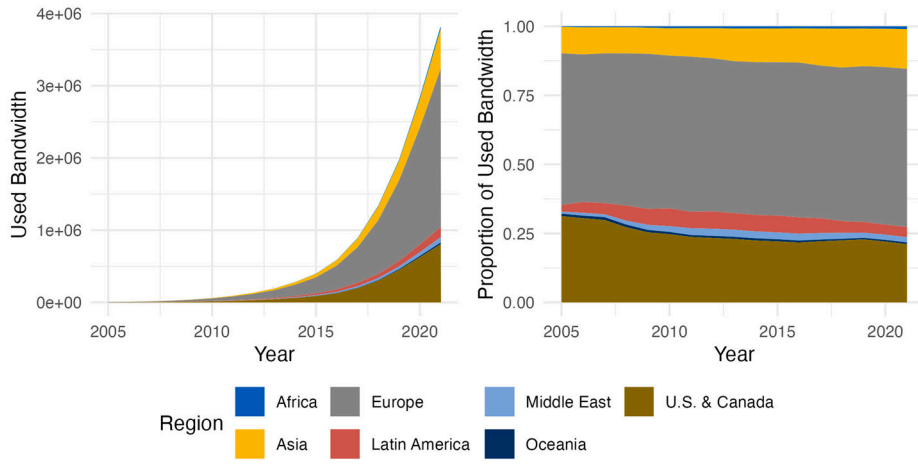
Table A.1 lists the largest 15 owners appearing in the TeleGeography data, where "largest" is determined by the number of cables in which each owner has a non-zero ownership share. It is worth noting that each of the owners appearing in this table is a major telecommunications firm.

Fig. A.1 provides a bit more context about the relative importance of the various global regions in terms of bandwidth demand. The left panel plots the total used bandwidth over time by region, and the right panel plots the proportion of global bandwidth represented by each region. It can be seen from the figures both that total bandwidth demand is growing exponentially in all regions, and that the relative importance

Table A.1
Cable Ownership: Largest Global Firms.

Firm Name	Number of Cables
Orange	26
Telecom Italia Sparkle	24
AT&T	23
Tata Communications	21
Arelion (formerly Telia Carrier)	19
C&W Networks	18
Vodafone	18
BT	17
Telkom Indonesia	16
Telstra	16
Verizon	16
Telekom Malaysia	14
XL Axiata	14
China Telecom	13
Singtel	13

Note: This table shows the largest 15 owners by cable-count in the TeleGeography data. The number of cables corresponds to cables where the owner has a strictly positive ownership share (which is not necessarily 100%).



Notes: The left panel plots used bandwidth over time, broken down by region. The right panel plots the proportion of global used bandwidth by region. Authors' calculations from TeleGeography data.

Fig. A.1. Bandwidth Demand by Region.

of the largest regions (Europe and North America) is only slowly declining.

Appendix B. International bandwidth demand growth rates

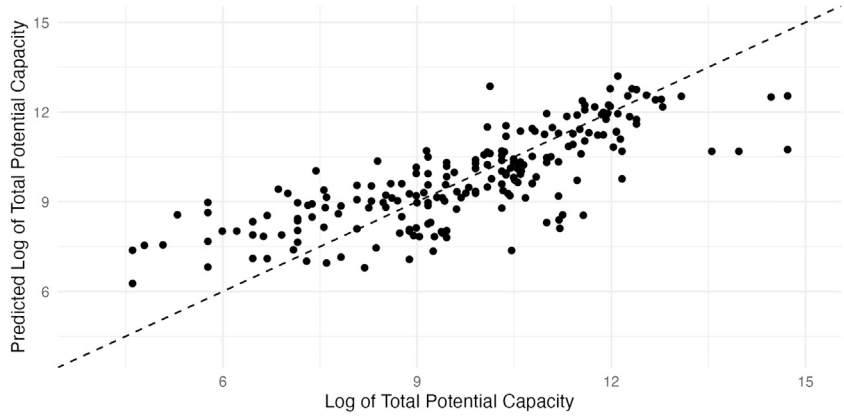
Further details of international bandwidth demand growth can be found in Table B.2; this table presents compound annual growth rates for specific market periods, routes, and buyer segments. The periods considered are 2005 to 2009, 2010 to 2014, 2015 to 2021, and the entire period of 2005 to 2021. Routes considered are the nine major subsea cable routes as well as the global total. Buyer segments are broken down into content providers, enterprise, internet backbone providers, and research and education networks.

This table shows that much of the international bandwidth growth observed over the period of the data, especially in the last 11 years, is driven primarily by content providers. Content providers have had the highest overall growth rate over the sample globally and on every route except for 1 (Europe-East Asia). In many cases the difference in growth rates is an order of magnitude, and it represents the difference in compound annual growth rate over the 16 year period of the sample.

Table B.2
International Bandwidth Demand Growth Rates.

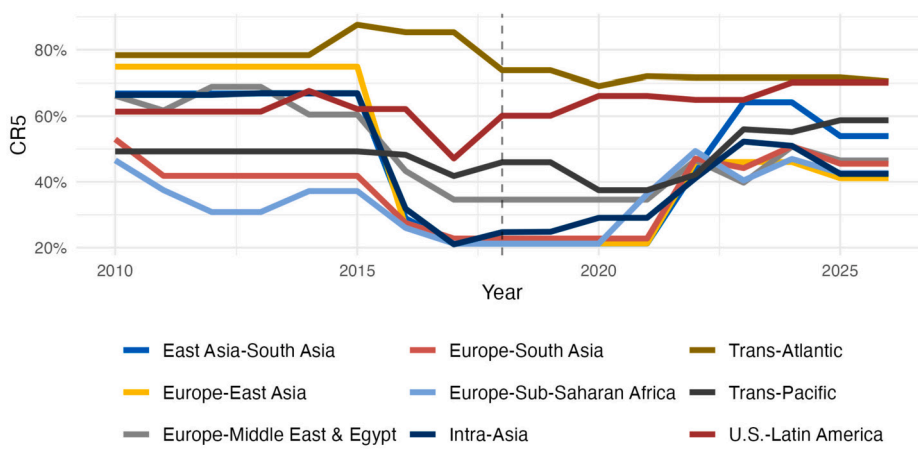
	Global Total	East Asia - South Asia	Europe - East Asia	Europe - Middle East & Egypt	Europe - South Asia	Europe - Sub-Saharan Africa	Intra-Asia	Trans-Atlantic	Trans-Pacific	U.S. - Latin America
					All Sources					
2005-09	57.6	93.5	165.1	61.7	88.3	88.3	53.8	43.4	47.9	91.5
2010-14	46.0	49.3	44.5	53.4	50.5	62.3	57.6	40.3	51.3	42.6
2015-21	45.8	66.7	42.9	42.9	60.6	55.5	43.4	51.6	43.9	31.8
2005-21	48.9	67.5	68.9	52.8	65.3	68.6	49.0	45.0	46.8	52.1
					Content Providers					
2005-09	76.9	–	–	–	–	–	354.9	35.4	58.4	223.2
2010-14	114.9	82.1	0	537.5	81.7	140.6	128.6	112.3	109.7	95.2
2015-21	63.6	95.5	165.5	108.5	135.1	111.1	54.0	63.4	56.7	61.9
2005-21	79.0	85.3	66.0	194.0	112.8	121.7	98.8	66.6	70.8	92.5
					Enterprise					
2005-09	47.3	74.8	150.2	33.8	92.0	70.5	42.5	31.3	32.4	83.5
2010-14	32.9	44.5	39.8	39.8	43.1	57.8	38.6	18.4	31.0	37.5
2015-21	25.1	49.6	33.5	32.0	37.8	41.0	23.5	16.6	21.9	21.5
2005-21	33.4	53.7	59.7	35.7	54.2	57.6	32.3	21.0	27.3	43.4
					Internet Backbone Providers					
2005-09	61.2	91.6	167.1	82.1	105.1	78.4	50.4	49.0	53.5	92.0
2010-14	36.4	47.0	46.1	55.6	49.3	64.5	41.0	20.1	38.4	41.4
2015-21	29.0	53.9	37.2	38.3	41.6	45.0	27.0	16.1	26.1	25.2
2005-21	40.1	61.0	66.9	56.7	61.2	63.8	37.0	26.0	37.1	48.7
					Research and Education Networks					
2005-09	27.6	–	–	32.1	15.4	531.7	52.5	40.6	21.8	27.5
2010-14	33.3	34.8	78.0	43.7	65.1	18.8	19.1	54.7	14.0	−6.4
2015-21	32.6	29.5	57.0	46.3	0.7	38.1	51.2	7.6	38.9	49.3
2005-21	29.9	46.4	54.7	37.4	13.8	77.5	35.3	28.4	22.8	47.2

Notes: Each cell contains the calculated compound annual growth rate for bandwidth demand for a given period (denoted on the left), subsea route, and type of purchaser. Global totals are shown in the first column. Totals for all sources are shown in the top panel; those for particular sources are shown in subsequent panels. Authors' calculations from TeleGeography's GRBS dataset.



Notes: This plot depicts a measure of fit of the regression of log potential capacity on cable length, year of service entry, and regions connected. The horizontal axis is the dependent variable used in the regression, and the vertical axis is the fitted value. The dashed line represents the 45-degree line, or line of perfect fit.

Fig. C.2. Capacity Regression Fit.



Notes: This figure plots the CR5, or the sum of the ownership share of predicted capacity of the five largest owners across time for each major route. The vertical line in 2018 indicates the point after which 95% of content provider entry occurred. Further details can be found in the text. Authors' calculations from TeleGeography data and reports; data from 2022 onward is TeleGeography's projected cable completions based on under-construction and announced cables.

Fig. C.3. Ownership Concentration, Predicted Capacity.

Appendix C. Concentration measures

To examine how recent trend of investment by content providers has affected concentration in cable ownership, we construct a measure of capacity ownership concentration at the year-route level, the CR5 ratio.[28] The CR5 ratio is simply the sum of the (ownership) shares of the largest five sellers in a given market. As capacity data is only available for about 50% of the cables in our sample, we present results using statistically *predicted capacity*.

To circumvent this issue, we regress (the log of) cable capacity on cable length as well as fixed effects denoting the cable's regions, the year of entry, and whether it is an intra-regional or inter-regional cable:

$$\log(C_i) = \beta_0 + \beta_1 KM_i + \beta_2 IntraRegion_i + \beta_3 InterRegion_i + \alpha_i + \rho_i + \epsilon_i \,. \tag{4}$$

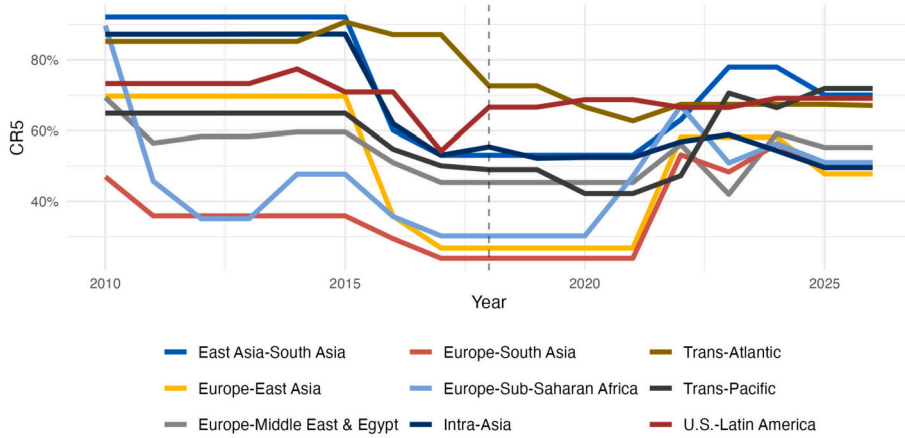
In Equation (4), C_i denotes the cable's capacity in Gbps, KM_i is the length of the cable in kilometers, $InterRegion_i$ and $IntraRegion_i$ are indicator variables equal to 1 if the cable has inter- and intra-regional landing points, respectively, and α_i and ρ_i are year-of-entry and region fixed effects. This regression yields an R^2 coefficient of 0.60 and an adjusted R^2 value of 0.51. Fig. C.2 illustrates the goodness of fit, plotting predicted values against the dependent variable for the estimation dataset.

For the subsequent analysis in this Appendix, we predict capacity using the coefficients estimated from Equation (4) and use predicted cable capacity for the entire universe of cables. This approach has the benefit of ensuring consistency across the data used, and the drawback of throwing out the genuine data in favor of a statistically generated substitute for a subset. Note also that we project these ratios forward in time, given the current slate of announced and under-construction cables: planned and under construction cables have entry dates ranging from the 2022 through 2026.[29]

In addition to the limitation presented by missing data on cable capacity, an additional limitation we face is that while the cable owner identities are fairly well populated, reported ownership shares

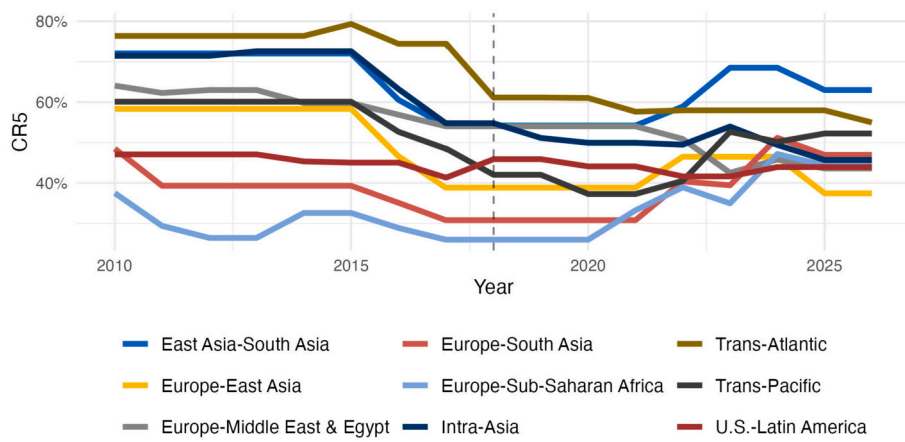
[28] Results using the Herfindahl–Hirschman Index are qualitatively similar.

[29] Recall the GBRS data at our disposal (described in Section 2.2 finishes in 2021.



Notes: This figure plots the CR5, or the sum of the ownership share of capacity (data or predicted) of the five largest owners across time for each major route. The vertical line in 2018 indicates the point after which 95% of content provider entry occurred. Authors' calculations from TeleGeography data and reports; data from 2022 onward is TeleGeography's projected cable completions based on under-construction and announced cables.

Fig. C.4. Ownership Concentration, Capacity (Data + Predicted).



Notes: This figure plots the CR5, or the sum of the ownership share of predicted fiber pairs of the five largest owners across time for each major route. The vertical line in 2018 indicates the point after which 95% of content provider entry occurred. Authors' calculations from TeleGeography data and reports; data from 2022 onward is TeleGeography's projected cable completions based on under-construction and announced cables.

Fig. C.5. Ownership Concentration, Predicted Fiber Pairs.

are mostly missing. Thus, to be able to proceed with an analysis of ownership-concentration, we assume equal ownership among all reported owners of the cable's residual ownership share (that is, the ownership share not directly reported). That is, for a cable i, with reported owners missing ownership share information $o \in \mathcal{O}$, and reported owners with ownership share information $u \in \mathcal{U}$, with shares $\{s_u\}$ we estimate:

$$\hat{s}_o = \frac{1 - \sum_{\mathcal{U}} s_u}{|\mathcal{O}|}.$$

Fig. C.3 plots the CR5 ratios in predicted capacity by route over time. While there is some important route-level heterogeneity, the overall pattern is one of decreasing concentration over the period of 2015-2018 with the entry of new cables and owners in the market, followed by an increase in predicted concentration from 2021 and on. This otherwise striking pattern is noticeably absent for the Trans-Atlantic and U.S.-Latin America routes, and relatively subdued for the Trans-Pacific route. While on first glance the increase in predicted ownership concentration across many routes appears concerning, a closer examination reveals that it represents a return to the *status quo ex ante*. At the end of this section, we perform two robustness checks of this analysis, finding similar results in both cases to those presented here.

The findings of this analysis are murkier than those in Section 3, where increasing buyer concentration, driven largely by content providers, is clearly documented. In contrast, the story of supplier concentration is not so straightforward and no clear conclusions can be drawn regarding the role played by content providers.

The trends presented here may be an area of interest for policy makers, though we stress that the confidence of any conclusion that could be drawn from the present analysis is limited by the assumptions we make to impute missing data (for both ownership shares and cable capacities). Additionally, it is important to note that it appears from this analysis that the recent increase in ownership concentration does not exceed historical concentration levels. A final point worth noting is that Caoui and Steck (2024) document falling bandwidth prices in another TeleGeography dataset over the 2005-2021 period, which argues against increased market power being a major concern over this time-frame.

Robustness Checks

As a first robustness check to this exercise, we repeat the analysis with the alternative approach: we keep the genuine data where it exists, and use projected capacity only for those cables where we do not have

information on capacity.[30] The results of this analysis are presented in Fig. C.4. While slight differences are visible in the figure compared to that presented in the main body of the paper, they do not lead to an alternative interpretation of the trends.

As an additional robustness check, we consider the results if we use a different dependent variable: the predicted number of owned fiber pairs. After using a regression similar to that described above by Equation (4), replacing cable capacity with the number of fiber pairs carried in the cable (an alternative measure of capacity), we generate the predicted number of fiber pairs using the estimated coefficients. Fig. C.5 plots the CR5 ratios in predicted fiber pairs by route over time. While most routes are predicted to have an increased concentration of ownership of fiber pairs in the coming years, these increases in general do not rise to the level of concentration exhibited in the first years of our sample.

Data availability

The authors do not have permission to share data.

References

Anderson, Benjamin, O'Connor, Alan C., 2020. Economic impact of 2Africa. RTI Int. (November).

Anderson, Benjamin, Merker, Jonathan, Wagstaff, Jeremy, Brower, Alice, Lakhani, Rizwaan, O'connor, Alan C., 2021. Economic impact of meta's subsea cable investments in Europe. RTI Int. (December), 1–34.

Becker, Gary S., Carlton, Dennis W., Sider, Hal S., 2010. Net neutrality and consumer welfare. J. Compet. Law Econ. 6 (3), 497–519.

Besen, Stanley, Milgrom, Paul, Mitchell, Bridger, Srinagesh, Padmanabhan, 2001. Advances in routing technologies and Internet peering agreements. Am. Econ. Rev. 91 (2), 292–296. https://doi.org/10.1257/aer.91.2.292.

Bourreau, Marc, Kourandi, Frago, Valletti, Tommaso, 2015. Net neutrality with competing Internet platforms. J. Ind. Econ. 63 (1), 30–73. https://doi.org/10.1111/joie.12068.

Brake, Doug, 2019. Submarine cables: critical infrastructure for global communications. Inf. Technol. Innov. Found. 1 (April), 1–11.

Briglauer, Wolfgang, 2015. How EU sector-specific regulations and competition affect migration from old to new communications infrastructure: recent evidence from EU27 member states. J. Regul. Econ. 48 (2), 194–217. https://doi.org/10.1007/s11149-015-9283-1.

Briglauer, Wolfgang, Frubing, Stefan, Vogelsang, Ingo, 2014. The impact of alternative public policies on the deployment of new communications infrastructure-a survey. Rev. Network Econom. 13 (3), 227–270. https://doi.org/10.1515/rne-2015-0004.

Briglauer, Wolfgang, Cambini, Carlo, Grajek, Michał, 2018. Speeding up the Internet: regulation and investment in the European fiber optic infrastructure. Int. J. Ind. Organ. 61, 613–652. https://doi.org/10.1016/j.ijindorg.2018.01.006.

Briglauer, Wolfgang, Stocker, Volker, Whalley, Jason, 2020. Public policy targets in EU broadband markets: the role of technological neutrality. Telecommun. Policy 44 (5), 101908. https://doi.org/10.1016/j.telpol.2019.101908.

Briglauer, Wolfgang, Cambini, Carlo, Gugler, Klaus, Stocker, Volker, 2023. Net neutrality and high-speed broadband networks: evidence from OECD countries. Eur. J. Law Econ. 55 (3), 533–571. https://doi.org/10.1007/s10657-022-09754-5.

Broos, Sébastien, Gautier, Axel, 2017. The exclusion of competing one-way essential complements: implications for net neutrality. Int. J. Ind. Organ. 52, 358–392. https://doi.org/10.1016/j.ijindorg.2017.03.003.

Callaway, Brantly, Sant'Anna, Pedro H.C., 2021. Difference-in-differences with multiple time periods. In: Treatment Effect 1. J. Econom. 225 (2), 200–230. https://doi.org/10.1016/j.jeconom.2020.12.001.

Caoui, El Hadi, Steck, Andrew, 2024. Diversification, Market Entry, and the Global Internet Backbone. Working Paper.

Cariolle, Joël, da Piedade, Camille, 2023. Digital connectedness and exports upgrading: is sub-Saharan Africa catching up? World Econ. 46 (11), 3325–3344. https://doi.org/10.1111/twec.13501.

Cariolle, Joël, Houngbonon, Georges Vivien, Silue, Tarna, Strusani, Davide, 2024. The Impact of Submarine Cables on Internet Access Price, and the Role of Competition and Regulation. World Bank Policy Research Working Paper.

Chang, Hyunseok, Jamin, Sugih, Willinger, Walter, 2006. To peer or not to peer: modeling the evolution of the Internet's AS-level topology. In: Proceedings IEEE INFOCOM 2006. 25TH IEEE International Conference on Computer Communications, pp. 1–12.

Chen, M. Keith, Nalebuff, Barry, 2006. One-way essential complements. http://ssrn.com/abstract=937384http://ssrn.com/abstract=942049http://www.som.yale.edu/Faculty/keith.chen/.

Cheng, Hsing Kenneth, Bandyopadhyay, Subhajyoti, Guo, Hong, 2011. The debate on net neutrality: a policy perspective. Inf. Syst. Res. 22, 60–82. https://doi.org/10.1287/isre.l090.0257.

Chiang, I. Robert, Jhang-Li, Jhih Hua, 2014. Delivery consolidation and service competition among Internet service providers. J. Manag. Inf. Syst. 31 (3), 254–286. https://doi.org/10.1080/07421222.2014.995561.

Choi, Jay Pil, Kim, Byung-Cheol, 2010. Net neutrality and investment incentives. Rand J. Econ. 41, 446–471.

Choi, Jay Pil, Jeon, Doh Shin, Kim, Byung Cheol, 2015. Net neutrality, business models, and Internet interconnection. Am. Econ. J. Microecon. 7 (3), 104–141. https://doi.org/10.1257/mic.20130162.

Clark, David D., Lehr, William, Bauer, Steven, 2011. Interconnection in the Internet: the policy challenge. In: The 39th Research Conference on Communication, Information and Internet Policy, pp. 1–24. http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1992641.

Conte, Maddalena, Cotterlaz, Pierre, Mayer, Thierry, 2021. The CEPII Gravity Database. CEPII, Paris, France.

Coucheney, Pierre, Maillé, Patrick, Tuffin, Bruno, 2014. Network neutrality debate and ISP inter-relations: traffic exchange, revenue sharing, and disconnection threat. Netnomics 15 (3), 155–182. https://doi.org/10.1007/s11066-014-9090-3.

Crémer, Jacques, Rey, Patrick, Tirole, Jean, 2000. Connectivity in the commercial Internet. J. Ind. Econ. 48, 433–472.

Demirer, Mert, Jiménez-Hernández, Diego, Li, Dean, Peng, Sida, 2023. Data, Privacy Laws and Firm Production: Evidence from the GDPR.

Easley, Robert F., Guo, Hong, Krämer, Jan, 2018. From net neutrality to data neutrality: a techno-economic framework and research agenda. https://doi.org/10.1287/isre.2017.0740.

Economides, Nicholas, 2008. "Net neutrality," non-discrimination and digital distribution of content through the Internet. I/S: J. Law Policy Inf. Soc.

Economides, Nicholas, Hermalin, Benjamin E., 2012. The economics of network neutrality. Rand J. Econ. 43, 602–629.

Economides, Nicholas, Tåg, Joacim, 2012. Network neutrality on the Internet: a two-sided market analysis. Inf. Econ. Policy 24, 91–104. https://doi.org/10.1016/j.infoecopol.2012.01.001.

European Parliament, 2023. Regulation (EU) 2022/1925 on contestable and fair markets in the digital sector (Digital Markets Act). March, https://eur-lex.europa.eu/ [Online; posted 20-March-2023].

Fanou, Rodérick, Huffaker, Bradley, Mok, Ricky, Claffy, K.C., 2020. Unintended consequences: effects of submarine cable deployment on Internet routing. In: Sperotto, Anna, Dainotti, Alberto, Stiller, Burkhard (Eds.), Passive and Active Measurement. Springer International Publishing, Cham, pp. 211–227.

Farronato, Chiara, Fradkin, Andrey, MacKay, Alexander, 2023. Self-preferencing at Amazon: evidence from search rankings. AEA Pap. Proc. 113, 239–243. https://doi.org/10.2139/ssrn.4342393.

Faulhaber, Gerald R., 2011. Economics of net neutrality: a review. Commun. Converg. Rev. 3 (1), 53–64. http://news.heartland.org/sites/default/files/EconomicsOfNetNeutrality.pdf%5Cnhttp://papers.ssrn.com/sol3/papers.cfm?abstract_id=1894286.

Federal Communications Commission, 2015. FCC Record 15-119: A Comprehensive Compilation of Decisions, Reports, Public Notices, and Other Documents of the Federal Communications Commission of the United States.

Ford, George S., 2018. Regulation and investment in the U.S. telecommunications industry. Appl. Econ. 50 (56), 6073–6084. https://doi.org/10.1080/00036846.2018.1489115.

Gans, Joshua S., 2015. Weak versus strong net neutrality. J. Regul. Econ. 47 (2), 183–200. https://doi.org/10.1007/s11149-014-9266-7.

Gans, Joshua S., Katz, Michael L., 2016. Weak versus strong net neutrality: correction and clarification. J. Regul. Econ. 50 (1), 99–110. https://doi.org/10.1007/s11149-016-9305-7.

Garbade, Kenneth D., Silber, William L., 1978. Technology, communication and the performance of financial markets: 1840-1975. J. Finance 33, 819–832. https://doi.org/10.1111/j.1540-6261.1978.tb02023.x.

Gardiner, Bryan, 2008. Google's submarine cable plans get official. February, https://www.wired.com [Online; posted 25-February-2008].

George, Lisa, Waldfogel, Joel, 2003. Who affects whom in daily newspaper markets? J. Polit. Econ. 111 (4), 765–784.

Goetz, Daniel, 2019. Dynamic bargaining and size EEects in the broadband industry. www.netinst.org.

Grajek, Michał, Röller, Lars Hendrik, 2011. Regulation and investment in network industries: evidence from European telecoms. J. Law Econ. 55 (1), 189–216. https://doi.org/10.1086/661196.

Greenstein, Shane, 2015. How the Internet Became Commercial: Innovation, Privatization, and the Birth of a New Network. Princeton University Press.

Greenstein, Shane, Pan Fang, Tommy, 2020. Where the Cloud Rests: the Economy Geography of Data Centers. Harvard Business School Working Paper, No. 21-042.

Greenstein, Shane, Peitz, Martin, Valletti, Tommaso, 2016. Net neutrality: a fast lane to understanding the trade-offs. J. Econ. Perspect. 30, 127–150. https://doi.org/10.1257/jep.30.2.127.

Hau, Thorsten, Burghardt, Dirk, Brenner, Walter, 2011. Multihoming, content delivery networks, and the market for Internet connectivity. Telecommun. Policy 35 (6), 532–542. https://doi.org/10.1016/j.telpol.2011.04.002.

[30] We thank an anonymous referee for this suggestion.

Hazlett, Thomas W., Wright, Joshua D., 2017. The effect of regulation on broadband markets: evaluating the empirical evidence in the FCC's 2015 "open Internet" order. Rev. Ind. Organ. 50 (4), 487–507. https://doi.org/10.1007/s11151-016-9556-6.

Hermalin, Benjamin E., Katz, Michael L., 2007. The economics of product-line restrictions with an application to the network neutrality debate. Inf. Econ. Policy 19, 215–248. https://doi.org/10.1016/j.infoecopol.2007.04.001.

Hildebrandt, Christian, Wiewiorra, Lukas, 2024. The past, present, and future of (net) neutrality: a state of knowledge review and research agenda. J. Inf. Technol. 39 (1), 167–193. https://doi.org/10.1177/02683962231170891.

Hjort, Jonas, Poulsen, Jonas, 2019. The arrival of fast Internet and employment in Africa. Am. Econ. Rev. 109, 1032–1079. https://doi.org/10.1257/aer.20161385.

Hoag, Christopher, 2006. The Atlantic telegraph cable and capital market information flows. J. Econ. Hist. 66, 342–353. https://doi.org/10.1017/S0022050706000143.

Hooton, Christopher Alex, 2020. Testing the economics of the net neutrality debate. Telecommun. Policy 44 (5), 101869. https://doi.org/10.1016/j.telpol.2019.101869.

Imbruno, Michele, Cariolle, Joel, de Melo, Jaime, 2022. Digital Connectivity and Firm Participation in Foreign Markets: An Exporter-Based Bilateral Analysis.

Jamison, Mark A., 2019. Net neutrality policies and regulation in the United States. Rev. Network Econom. 17 (3), 151–173. https://doi.org/10.1515/rne-2018-0041.

Jeon, Jihye, Rysman, Marc, 2024. Investment and Usage of the Subsea Internet Cable. (August).

Kang, Karam, Xiao, Mo, 2023. Policy Deterrence: Strategic Investment in U.S. Broadband.

Kim, Soo Jin, 2020. Direct interconnection and investment incentives for content quality. Rev. Network Econom. 18 (3), 169–204. https://doi.org/10.1515/rne-2020-0013.

Krämer, Jan, Wiewiorra, Lukas, 2012. Network neutrality and congestion sensitive content providers: implications for content variety, broadband investment, and regulation. Inf. Syst. Res. 23, 1303–1321. https://doi.org/10.1287/isre.1120.0420.

Krämer, Jan, Wiewiorra, Lukas, Weinhardt, Christof, 2013. Net neutrality: a progress report. Telecommun. Policy 37 (9), 794–813. https://doi.org/10.1016/j.telpol.2012.08.005.

Laffont, Jean-Jacques, Marcus, Scott, Rey, Patrick, Tirole, Jean, 2001. Internet peering. Am. Econ. Rev. 91, 287–291.

Laffont, Jean-Jacques, Marcus, Scott, Rey, Patrick, Tirole, Jean, 2003. Internet interconnection and the off-net-cost pricing principle. Rand J. Econ. 34, 370–390.

Layton, Roslyn, 2017. Which Open Internet Framework Is Best for Mobile App Innovation? An Empirical Inquiry of Net Neutrality Rules Around the World. Aalborg Universitetsforlag.

Lee, Daeho, Kim, Yong Hwa, 2014. Empirical evidence of network neutrality - the incentives for discrimination. Inf. Econ. Policy 29, 1–9. https://doi.org/10.1016/j.infoecopol.2014.07.001.

Lee, Robin S., Wu, Tim, 2009. Subsidizing creativity through network design: zero-pricing and net neutrality. J. Econ. Perspect. 23.

Little, Iain, Wright, Julian, 2000. Peering and settlement in the Internet: an economic analysis. J. Regul. Econ. 18 (2), 151–173. https://doi.org/10.1023/A:1008120618665.

Malone, Jacob B., Nevo, Aviv, Nolan, Zachary, Williams, Jonathan W., 2023. Is OTT video a substitute for TV? Policy insights from cord-cutting. Rev. Econ. Stat. 105 (6), 1615–1623. https://doi.org/10.1162/rest_a_01132.

McManus, Brian, Nevo, Aviv, Nolan, Zachary, Williams, Jonathan W., 2023. The steering incentives of gatekeepers in the telecommunications industry. https://doi.org/10.2139/ssrn.4203065.

McManus, Brian, Nevo, Aviv, Nolan, Zachary, Williams, Jonathan W., 2024. Does Pricing of Internet Usage Steer Consumers or Meter Usage? Evidence from a Pricing Experiment, pp. 1–46.

Musacchio, John, Schwartz, Galina, Walrand, Jean, 2009. A two-sided market analysis of provider investment incentives with an application to the net-neutrality issue. Rev. Network Econom. 8 (1). https://doi.org/10.2202/1446-9022.1168.

Nikkhah, Ali, Jordan, Scott, 2023. Analysis of the requirements of settlement-free interconnection policies. IEEE Trans. Netw. Serv. Manag. 20 (4), 4028–4046. https://doi.org/10.1109/TNSM.2023.3254367.

Njoroge, Paul, Ozdaglar, Asuman, Stier-Moses, Nicolá E., Weintraub, Gabriel Y., 2013. Investment in two-sided markets and the net neutrality debate. Rev. Network Econom. 12 (4), 355–402. https://doi.org/10.1515/rne-2012-0017.

OECD, 2014. The Internet in transition: the state of the transition to IPv6 in today's Internet and measures to support the continued use of IPv4. (234). https://doi.org/10.1787/5jz5sq5d7cq2-en.

Reimers, Imke, Waldfogel, Joel, 2023. A framework for detection, measurement, and welfare analysis of platform bias. https://doi.org/10.2139/ssrn.4596066.

Schuett, Florian, 2010. Network neutrality: a survey of the economic literature. Rev. Network Econom. 9. https://doi.org/10.2202/1446-9022.1224.

Simione, Felix, Li, Yiruo, 2021. The macroeconomic impacts of digitalization in Sub-Saharan Africa: evidence from submarine cables. IMF Work. Pap. 2021 (110), 1. https://doi.org/10.5089/9781513582542.001.

Skiti, Tedi, 2020. Strategic technology adoption and entry deterrence in broadband. Ind. Corp. Change 29 (3), 713–729. https://doi.org/10.1093/ICC/DTZ057.

Steinwender, Claudia, 2018. Real effects of information frictions: when the states and the kingdom became united. Am. Econ. Rev. 108, 657–696. https://doi.org/10.1257/aer.20150681.

Sunak, Rishi, 2017. Undersea cables indispensable, insecure. Policy Exch.

Swinhoe, Dan, 2021. What is a subsea cable? August, datacenterdynamics.com [Online; posted 26-August-2021].

Tan, Marvin, 2023. Content providers are still hungry for bandwidth. July, https://blog.telegeography.com [Online; posted 20-July-2023].

The Guardian, 2021. Major Internet outage 'shows infrastructure needs urgent fixing'. June, theguardian.com [Online; posted 11-June-2021].

The Register, 2021. 'It's time to decentralize the Internet. Again: what was distributed is now centralized by Google, Facebook, etc. August, theregister.com [Online; posted 11-August-2021].

The Register, 2022. Big Tech shrank the Internet while growing its own power. May, theregister.com [Online; posted 11-May-2022].

Waldfogel, Joel, 2003. Preference externalities: an empirical study of who benefits whom in differentiated-product markets. Rand J. Econ. 34 (3), 557. https://doi.org/10.2307/1593746.

Wang, Xin, Xu, Yinlong, Ma, Richard T.B., 2021. Paid peering, settlement-free peering, or both? IEEE/ACM Trans. Netw. 29 (2), 585–594. https://doi.org/10.1109/TNET.2020.3045220.

Wilson, Kyle, 2016. Does public competition crowd out private investment? Evidence from municipal provision of Internet access. SSRN Electron. J., 1–51. https://doi.org/10.2139/ssrn.2848569.

Wilson, Kyle, 2023a. Investment, subsidies, and universal service: broadband Internet in the United States. Rev. Network Econom. 22 (1), 1–25. https://doi.org/10.1515/rne-2023-0066.

Wilson, Kyle, 2023b. Local competition, multimarket contact, and product quality: evidence from Internet service provision. Rev. Ind. Organ. https://doi.org/10.1007/s11151-023-09928-8.

Wilson, Kyle, Xiao, Mo, Orazem, Peter F., 2021. Entry threat, entry delay, and Internet speed: the timing of the U.S. broadband rollout. J. Econ. Manag. Strategy 30 (1), 3–44. https://doi.org/10.1111/jems.12407.